

BARC YT SERIES 2025

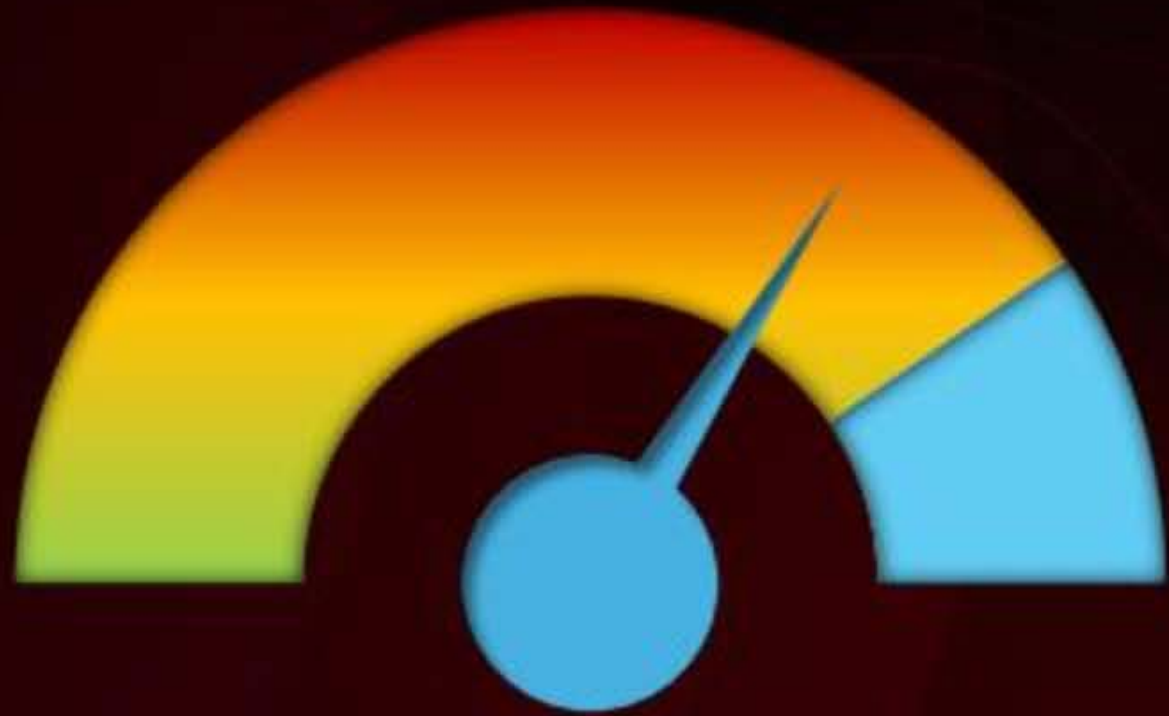
ELECTRICAL ENGINEERING
ELECTRONICS AND COMMUNICATION ENGINEERING

Digital Electronics

BARC PRACTICE
SESSION



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**TOPICS
TO BE COVERED**

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Question Series

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#Q. The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written as F^D , is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

n -variables $\rightarrow$ ✓ $N = 2^n$

$M = 2^N = 2^{2^n}$

$S_D = 2^{2^{n-1}}$

$NIS_D = (2^{2^n} - 2^{2^{n-1}})$

(D)

- A $2n$
- B 2^{n-1}
- C 2^{2^n}
- ☒ D $2^{2^{n-1}}$

#Q. If w, x, y, z are Boolean variables, then which one of the following is **INCORRECT**?

- A** $wx + w(x + y) + \underbrace{x(x + y)}_x = x + wy$ ✓✓
- B** $\overline{w\bar{x}(y + \bar{z}) + \bar{w}x} = \bar{w} + x + \bar{y}z \rightarrow \overline{w\bar{x}} + \bar{y}z + \bar{w}x$
 $\bar{w} + x + \bar{y}z + \bar{w}x = \bar{w} + x + \bar{y}z$
- C** ✓ $(\cancel{w\bar{x}(y + x\bar{z})} + \bar{w}\bar{x})y = x\bar{y}$ ✓
 $\Delta (\bar{w}\bar{x}y + \bar{w}\bar{x})y$
 $\Rightarrow \bar{x}y[(\bar{w} \cdot y) + \bar{w}]$
 $\bar{x}y[\bar{w} + y] \rightarrow \bar{x}y$
 $\bar{x} \cdot y[y + \bar{w}] = \bar{x}y$
- D** $(w + y)(\underline{w}xy + \underline{w}yz) = wxy + wyz$
 $(w + y)w(xy + yz)$
 $w(xy + yz) = wxy + wyz$
- $wx + wy + x = x + wy$
- $P \cdot [P + \text{anything}] = P$

#Q. The binary operator $\neq$ is defined by the following truth table

P	Q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

$$p \neq q = p \oplus q$$

Which one of the following is true about the binary operator $\neq$?

- ☒ **A** Both commutative and associative ✓
- ☐ **B** Commutative but not associative
- ☐ **C** Not commutative but associative
- ☐ **D** Neither commutative nor associative

#Q. Consider the Boolean function, $F(w, x, y, z) = wy + xy + \bar{w}xyz + \bar{w}\bar{x}y + xz + \bar{x}\bar{y}\bar{z}$. Which one of the following is the complete set of essential prime implicants?

- A** $w, y, xz, \bar{x}\bar{z}$
- B** w, y, xz
- C** $y, \bar{x}\bar{y}\bar{z}$
- D** $y, xz, \bar{x}\bar{z}$

#Q. The Boolean expression
 $(X + Y)(X + \bar{Y}) + \overline{(\bar{X}\bar{Y} + \bar{X})}$ simplifies to

A X

B Y

C XY

D $X + Y$

$$[X + Y \cdot \bar{Y}]$$

$$X + X = X$$

#Q. In the sum of products function $f(X, Y, Z) = \Sigma(2, 3, 4, 5)$, the prime implicants are

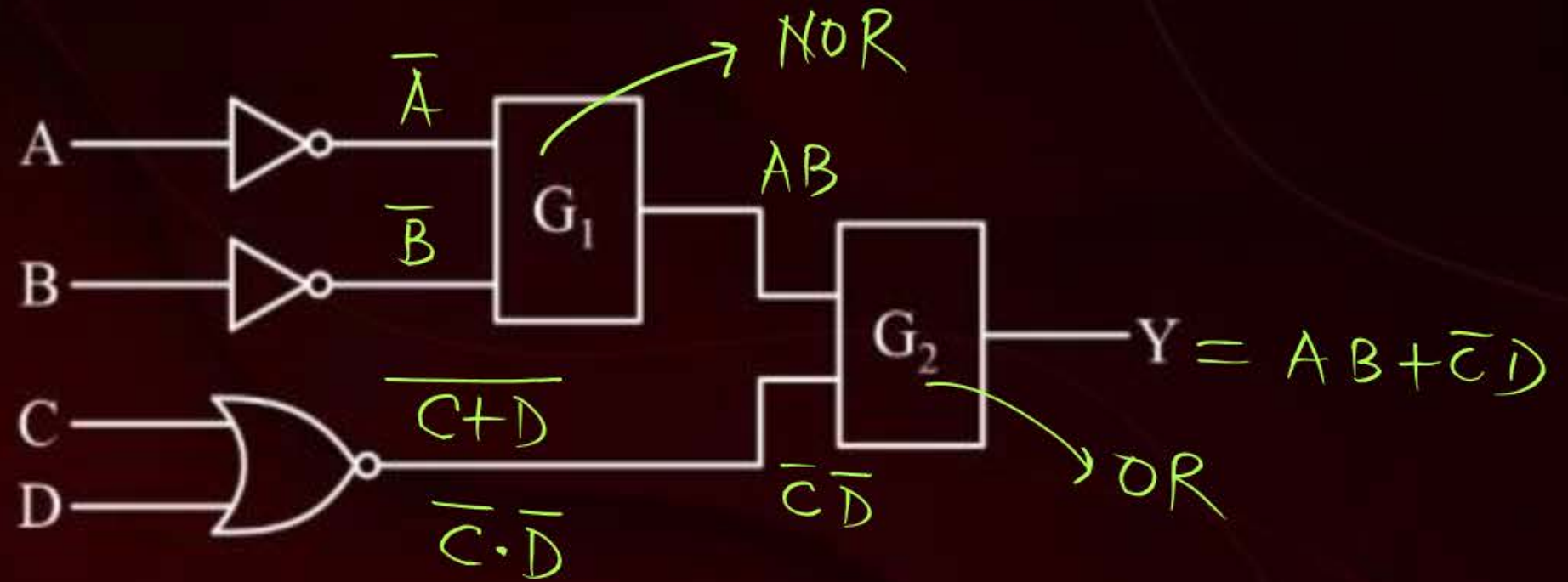
- A** $\bar{X}Y, X\bar{Y}$
- B** $\bar{X}Y, X\bar{Y}\bar{Z}, X\bar{Y}Z$
- C** $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}$
- D** $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}\bar{Z}, X\bar{Y}X$

#Q. Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator.
Which one of the following must always be TRUE?

- A** $x_1 x_2 x_3 x_4 = 0$
- B** $x_1 x_3 + x_2 = 0$
- C** $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
- D** $x_1 + x_2 + x_3 + x_4 = 0$

#Q. In the figure shown, the output Y is required to be $Y = AB + \bar{C}\bar{D}$. The gates G_1 and G_2 must be,

- A** ✓ NOR, OR
- B** OR, NAND ✗
- C** NAND, OR
- D** AND, NAND ✗



#Q. P, Q, and R are the decimal integers corresponding to the 4-bit binary number 1100 considered in signed magnitude, 1's complement, and 2's complement representations, respectively. The 6-bit 2's complement representation of $(P + Q + R)$ is

- ☒ **A** 110101
- ☐ **B** 110010
- ☐ **C** 111101
- ☐ **D** 111001

$$1100 \longrightarrow P = -(4) \text{ sign mag.}$$

$$Q = -(3)$$

$$\downarrow 1's$$

$$(0011)_2$$

$$R = -8 + 4 \times 1 + 0 + 0 = -4$$

$$(P+Q+R) = (-11)_2 \longrightarrow 5 \text{ bit}$$

$$\Rightarrow -(2^{n-1}) \text{ to } +(2^{n-1}-1)$$

$$(-11)_2 \rightarrow \overset{2^4}{[10101]}$$

$$\begin{array}{r} 110101 \\ \hline \end{array}$$

$$(-23)_2 \rightarrow (101001)_2 \checkmark \checkmark$$

#Q. The two numbers represented in signed 2's complement form are $P = 11101101$ and $Q = 11100110$. If Q is subtracted from P, the value obtained in signed 2's complement form is

A 100000111 ~~X~~

B 00000111

C 11111001 ~~X~~

D 111111001 ~~X~~

$$(P - Q) = (7)_{10}$$

$$P = (111\underline{0}1101)_2 = (101101)_2 = (-19)_{10}$$

$$Q = 11100110 = (100110)_2 = (-26)_{10}$$

$$Q - P = (-7)_{10} = (1001) = (11111001)_2$$

#Q. In 16-bit 2's complement representation, the decimal number -28 is:

- A** 1111 1111 0001 1100 ✗
- B** 0000 0000 1110 0100 ✗
- C** ✓ 1111 1111 1110 0100
- D** ✗ 1000 0000 1110 0100

$$(-28)_{10} = (100100)_2$$

#Q. Consider $Z = X - Y$, where X , Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

- A** n bits
- B** $n - 1$ bits
- C** $n + 1$ bits
- D** $n + 2$ bits

$$\underline{n=4 \text{ bit}} \longrightarrow -(2^{n-1}-1) \text{ to } +(2^{n-1}-1)$$

$$-7 \text{ to } +7$$

$$\left. \begin{array}{l} \underline{\underline{5 \text{ bit}}} \\ (n+1) \text{ bit} \end{array} \right\} \begin{array}{l} (X-Y)_{\min} = -7 - 7 = (-14) \\ (X-Y)_{\max} = 7 - (-7) = (14) \end{array}$$

#Q. The smallest integer that can be represented by an 8-bit number in 2's complement form is

- A** - 256
- B** - 128
- C** - 127
- D** 0

$$-(2^7) \text{ to } +(2^7 - 1)$$

$$(-128 \text{ to } +127) \checkmark \checkmark$$

#Q. When two 8-bit numbers $A_7 \dots A_0$ and $B_7 \dots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a ripple-carry adder, the sum bits obtained are $S_7 \dots S_0$ and the carry bits are $C_7 \dots C_0$.
An overflow is said to have occurred if

- A** The carry bit C_7 is 1
- B** All the carry bits ($C_7, \dots, C_0$) are 1
- C** $(A_7 \cdot B_7 \cdot \bar{S}_7 + \bar{A}_7 \cdot \bar{B}_7 \cdot S_7)$ is 1.
- D** $(A_0 \cdot B_0 \cdot \bar{S}_0 + \bar{A}_0 \cdot \bar{B}_0 \cdot S_0)$ is 1.

#Q. Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121}_r = 11_r$ is/are

- A** decimal 10
- B** decimal 11
- C** decimal 10 and 11
- D** any value > 2

$$(121)_r \rightarrow r \geq 3 \text{ or } \underline{\underline{r > 2}}$$

$$\sqrt{(121)_r} = (11)_r$$

$$(r^2 + 2r + 1)^{1/2} = (r + 1)_r$$

$$r^2 + 2r + 1 = (r^2 + 2r + 1) \rightarrow \text{infinite solutions}$$

$$\rightarrow r > 2$$

$$\sqrt{(625)_n} = (25)_n \quad \underline{n=10}$$

$n \geq 7 \checkmark$

$$(6n^2 + 2n + 5)_{10}^{1/2} = (2n + 5)$$

$$6n^2 + 2n + 5 = 4n^2 + 25 + 20n$$

$$2n^2 - 18n - 20 = 0$$

$$2n^2 - 20n + 2n - 20 = 0$$

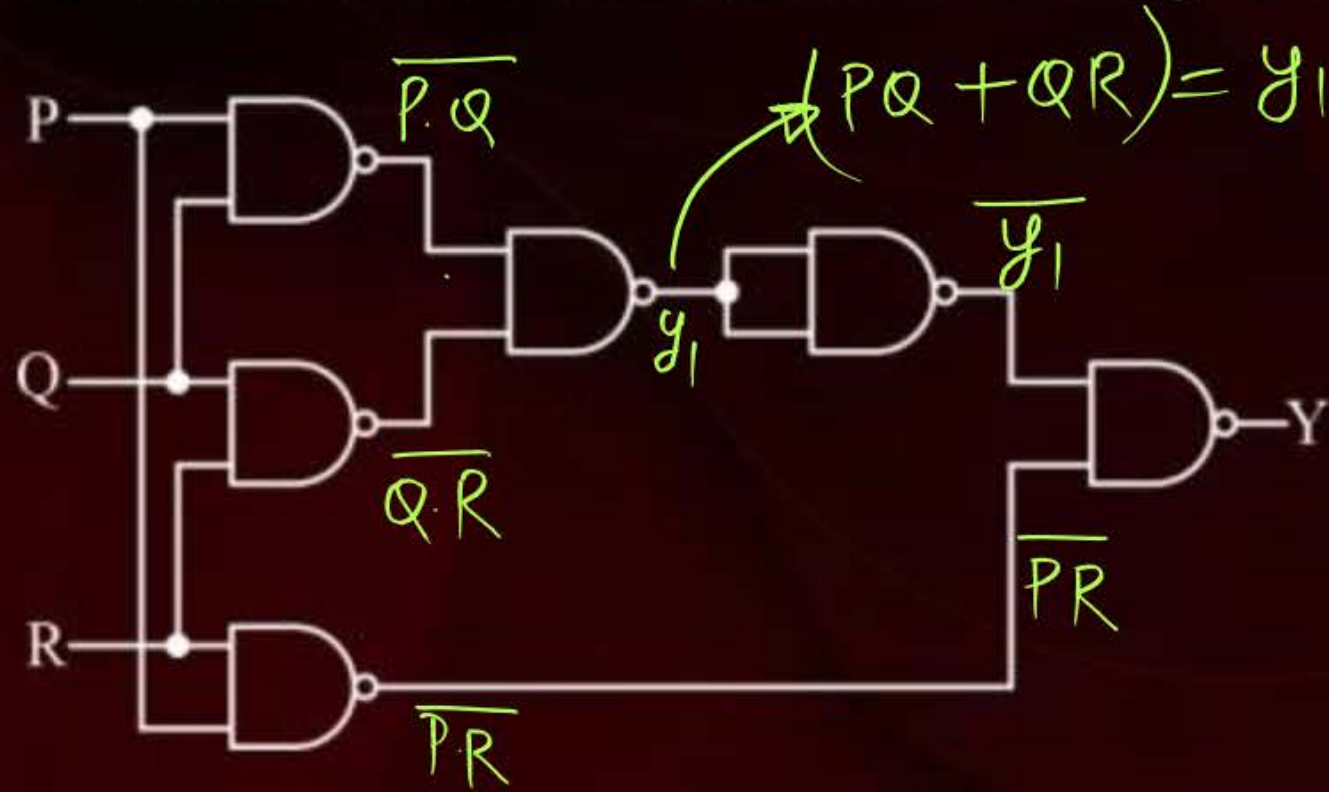
$$2n(n - 10) + 2(n - 10) = 0$$

$$(2n + 2)(n - 10) = 0$$

$$n = -1, 10$$

X $\checkmark$

#Q. The output Y in the circuit below is always '1' when



$$Y = Y_1 + P.R$$

$$= (P.Q + Q.R + P.R)$$

majority I/P CKT

$\downarrow$
 $\left\{ \begin{array}{l} 1 \rightarrow \text{majority are '1'} \\ 0 \rightarrow \text{majority are '0'} \end{array} \right.$

A two or more of the inputs P, Q, R are '0' ~~X~~

B two or more of the inputs P, Q, R are '1' ✓

C any odd number of the inputs P, Q, R is '0'

D any odd number of the inputs P, Q, R is '1'

#Q. The boolean expression:
 $\overline{(a + \bar{b} + c + \bar{d}) + (b + \bar{c})}$ simplifies to

- A** 1
- B** $\overline{a.b}$
- C** $A.b$
- D** 0

#Q. Which of the following logic circuits is a realization of the function F whose Karnaugh map is shown in figure?

C \ AB	AB			
	00	01	11	10
0	1	1		
1		1	1	

A



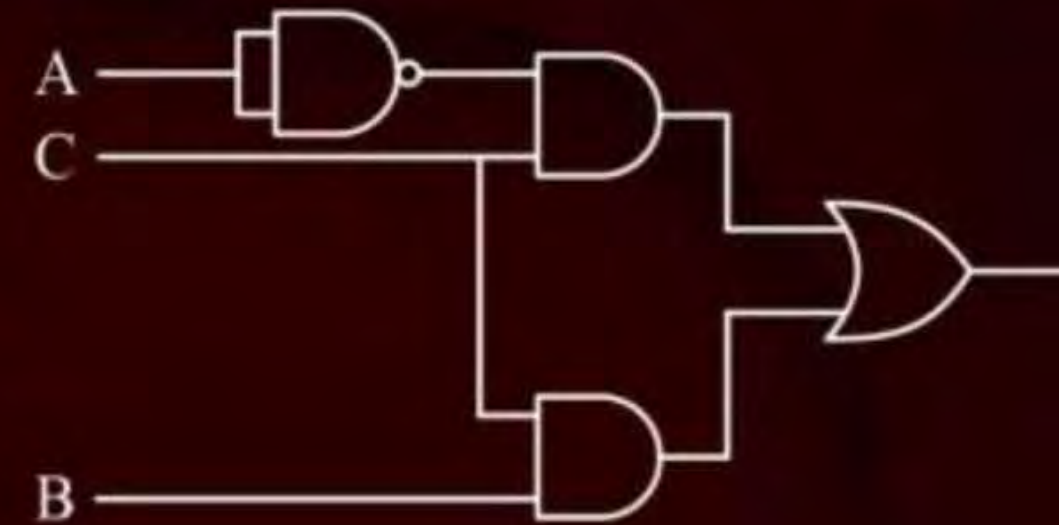
B



C



D



#Q. Which one of the following is NOT a valid identity?

- A** $(x + y) \oplus z = x \oplus (y + z)$ ✗
- B** $(x \oplus y) \oplus z = x \oplus (y \oplus z)$ ✓
- C** $x \oplus y = x + y$, if $xy = 0$ ✓
- D** $x \oplus y = \overline{(xy + x'y')}$ ✓
 $\overline{xy}$

$$x \oplus y = z \rightarrow x \oplus z = y$$

$$x \oplus (y + z) \neq (x \oplus y) + (x \oplus z)$$

$$x \cdot y \rightarrow (0, 0), (0, 1), (1, 0)$$

	0	1	1
$x \oplus y$	0	1	1
$x + y$	0	1	1

#Q. Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q:

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

$$1 \oplus P \oplus Q = \bar{P} \oplus Q = P \odot Q$$

$$P \oplus P \oplus P = P$$

$$Q \oplus Q \oplus Q = Q$$

A $P + Q$

B $(P + Q)'$

C $P \oplus Q$

D $(P \oplus Q)'$

#Q. What is the Boolean expression for the output f of the combinational logic circuit of NOR gates given below? ✓✓

- A** $\overline{Q + R}$
- B** $\overline{P + Q}$
- C** $\overline{P + Q}$
- D** $\overline{P + Q + R}$



#Q. A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?

- A** EX-NOR
- B** implication, negation
- C** OR, negation
- D** NAND

#Q. In the 4×1 multiplexer, the output F is given by $F = A \oplus B$. Find the required input ' $I_3 I_2 I_1 I_0$ '

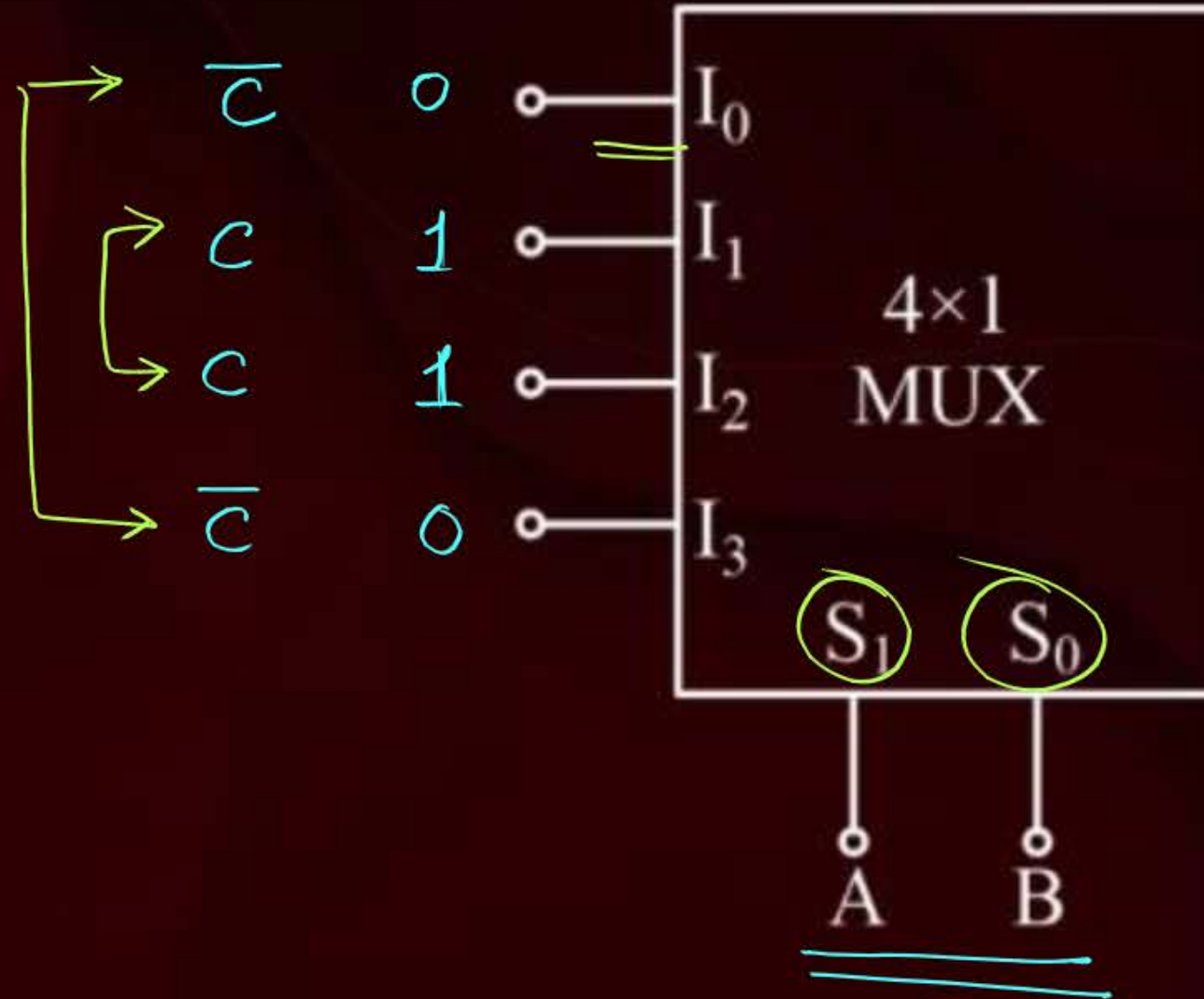
$$F(A, B) = \sum(1, 2) = \pi(0, 3)$$

A 1010

B 0110

C 1000

D 1110



$$\begin{aligned} F &= \overline{C} \oplus S_1 \oplus S_0 \\ &= \overline{C} \oplus A \oplus B \\ &= C \odot A \oplus B \\ &= C \oplus A \odot B \\ &= A \oplus B \odot C \\ &= A \odot B \oplus C \end{aligned}$$

#Q. The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The number of combinations for which the output is logic 1, is

- A** 4
- B** 6
- C** 8
- D** 10

4 bit
= 16 Combination

A 2 bit	B 2 bit
0	0
1	
2	
3	3

$$N_1(A > B) = 0 + 1 + 2 + 3 = 6$$

$$N_2(A < B) = 0 + 1 + 2 + 3 = 6$$

$$N_3(A = B) = 4$$

#Q. A 4 :1 multiplexer is to be used for generating the output carry of a full adder. A and B are the bits to be added while C_{in} is the input carry C_{out} is the output carry. A and B are to be used as the select bits with A being the more significant select bit.

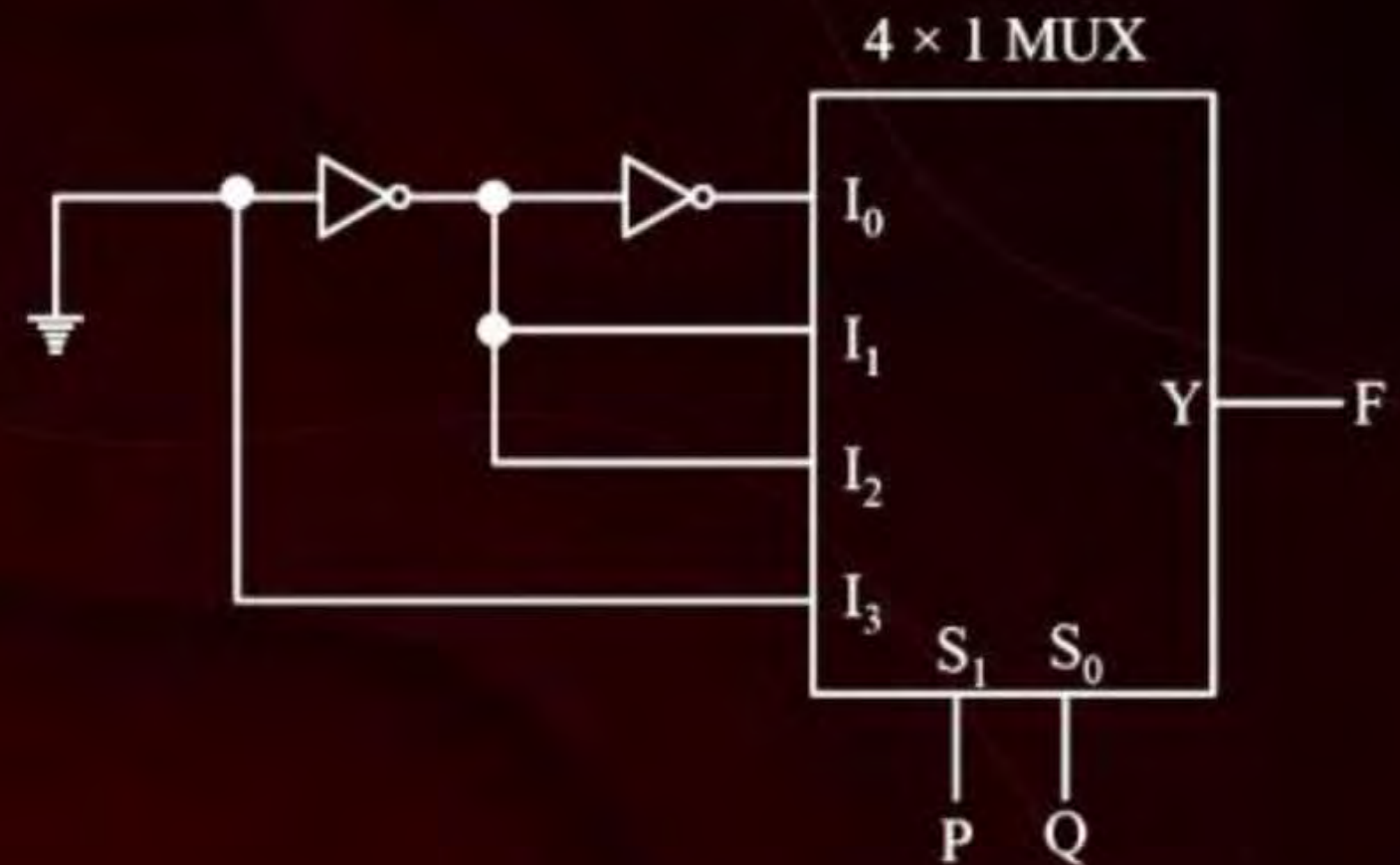
Which one of the following statements correctly describes the choice of signals to be connected to the inputs I_0 , I_1 , I_2 and I_3 so that the output is C_{out} ?

- A** $I_0 = 0, I_1 = C_{in}, I_2 = C_{in}$ and $I_3 = 1$
- B** $I_0 = 1, I_1 = C_{in}, I_2 = C_{in}$ and $I_3 = 1$
- C** $I_0 = C_{in}, I_1 = 0, I_2 = 1$ and $I_3 = C_{in}$
- D** $I_0 = 0, I_1 = C_{in}, I_2 = 1$ and $I_3 = C_{in}$



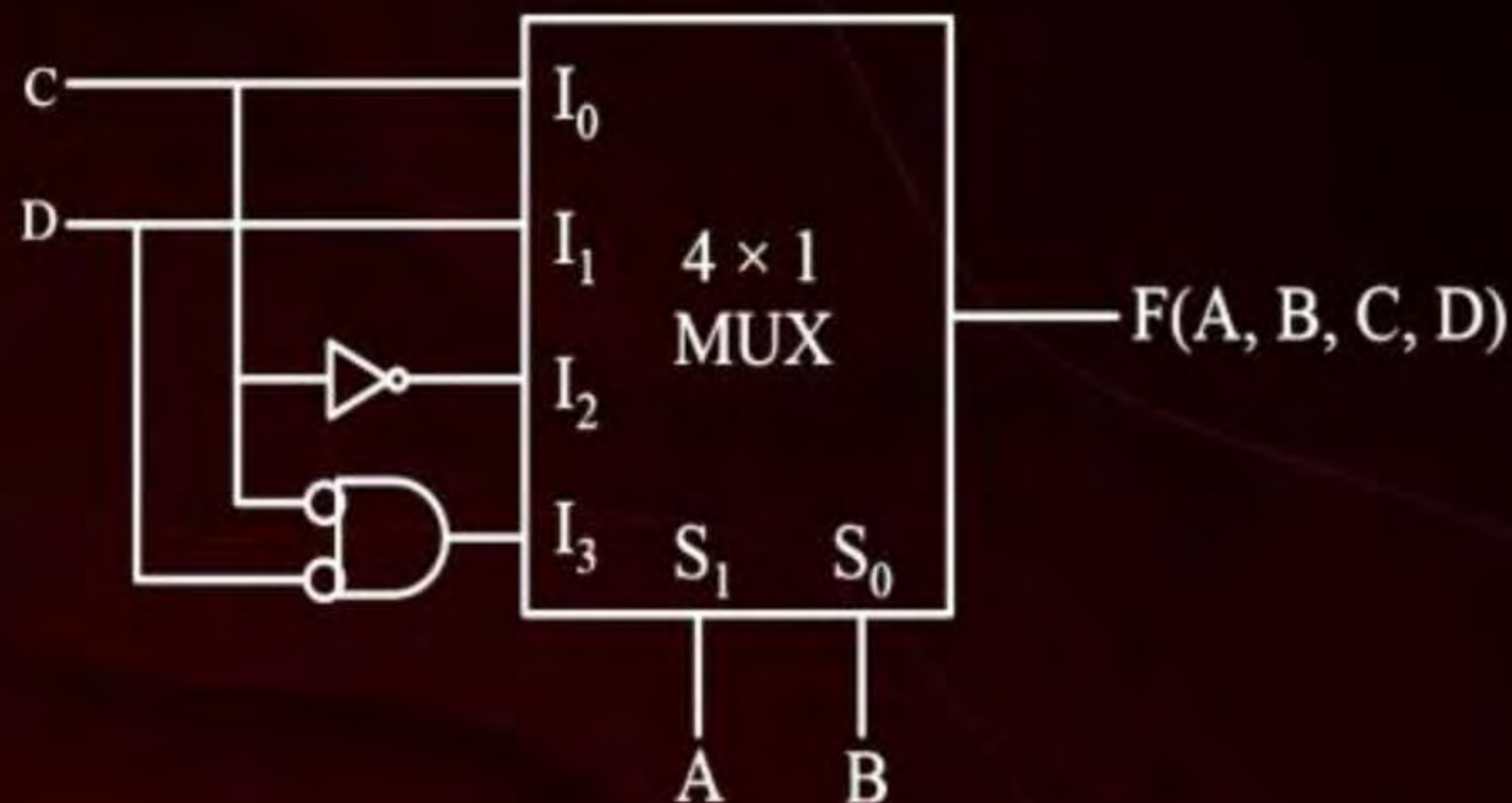
#Q. The logic function implemented by the circuit below is (ground implies a logic '0')

- A** $F = \text{AND}(P, Q)$
- B** $F = \text{OR}(P, Q)$
- C** $F = \text{XNOR}(P, Q)$
- D** $F = \text{XOR}(P, Q)$



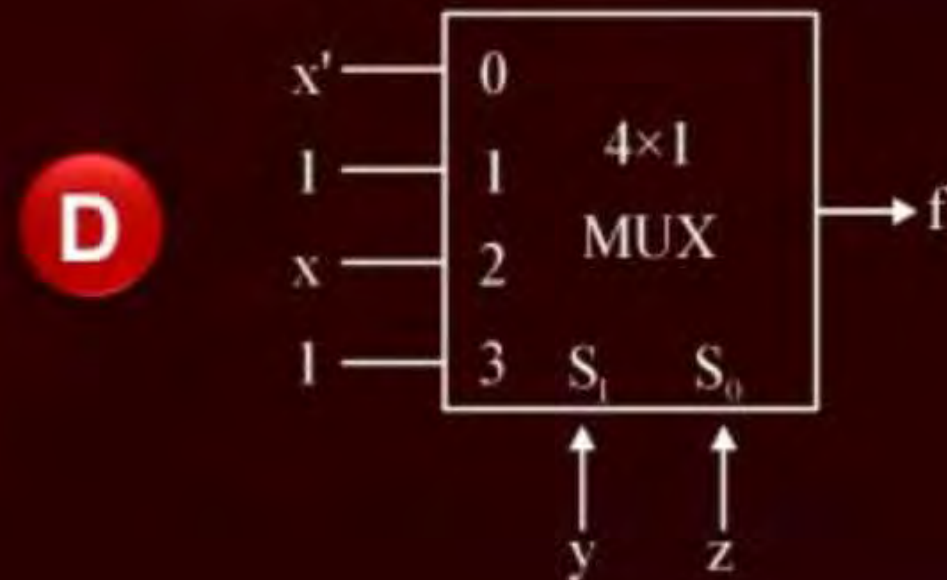
#Q. The Boolean function realized by the logic circuit shown is

- A** $F = \sum m (0, 1, 3, 5, 9, 10, 14)$
- B** $F = \sum m (2, 3, 5, 7, 8, 12, 13)$
- C** $F = \sum m (1, 2, 4, 5, 11, 14, 15)$
- D** $F = \sum m (2, 3, 5, 7, 8, 9, 12)$



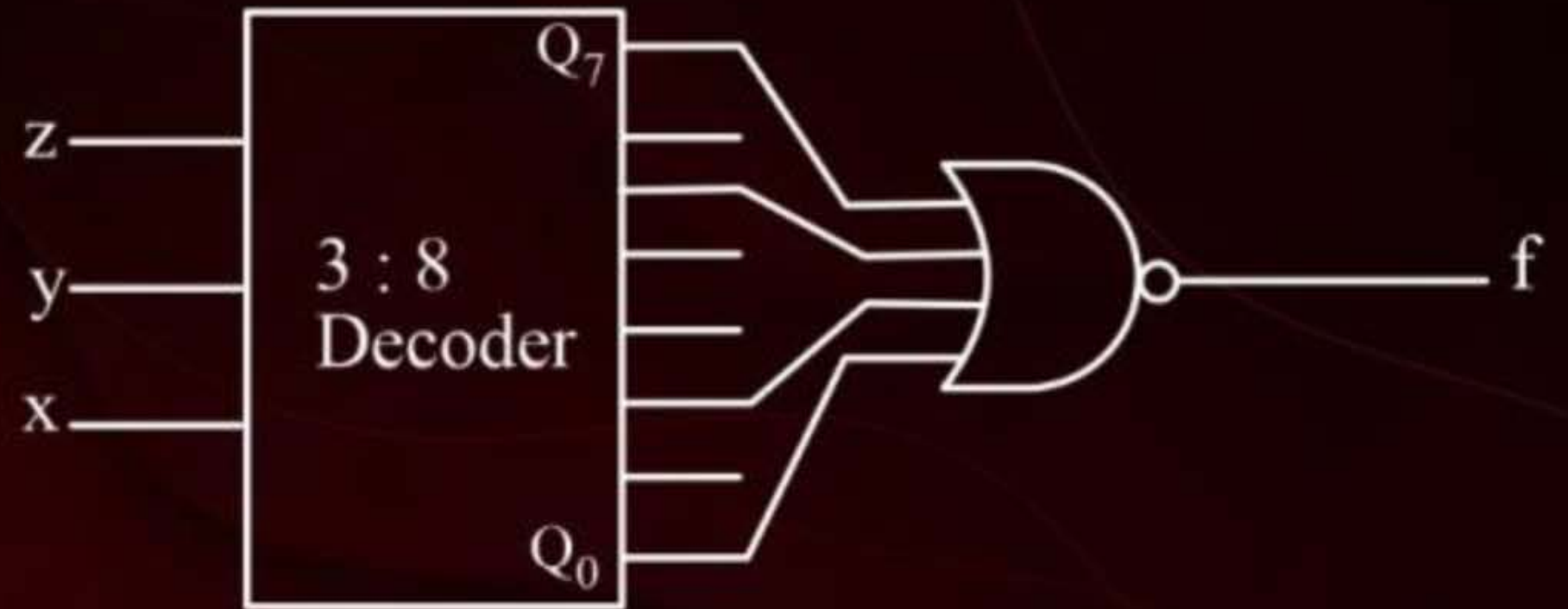
#Q. Which one of the following circuits implements the Boolean function given below?

$$f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6 \text{ where } m_t \text{ is the } t^{\text{th}} \text{ minterm.}$$



#Q. What Boolean function does the circuit below realize?

- A** $xz + \bar{x}\bar{z}$
- B** $x\bar{z} + \bar{x}z$
- C** $\bar{x}\bar{y} + yz$
- D** $xy + \bar{y}\bar{z}$

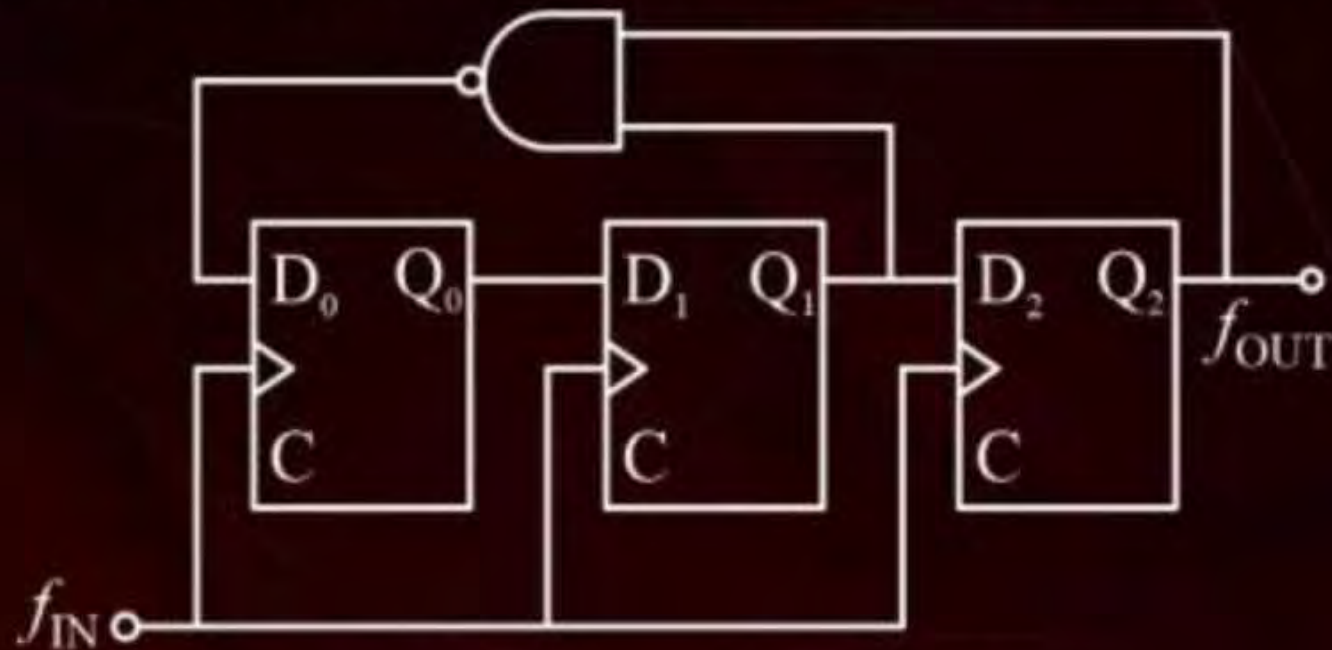


#Q. In the following truth table, $V = 1$ if and only if the input is valid. What function does the truth table represent?
What function does the truth table represent?

- A** ✓ Priority encoder
- B** Decoder
- C** Multiplexer
- D** Demultiplexer

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

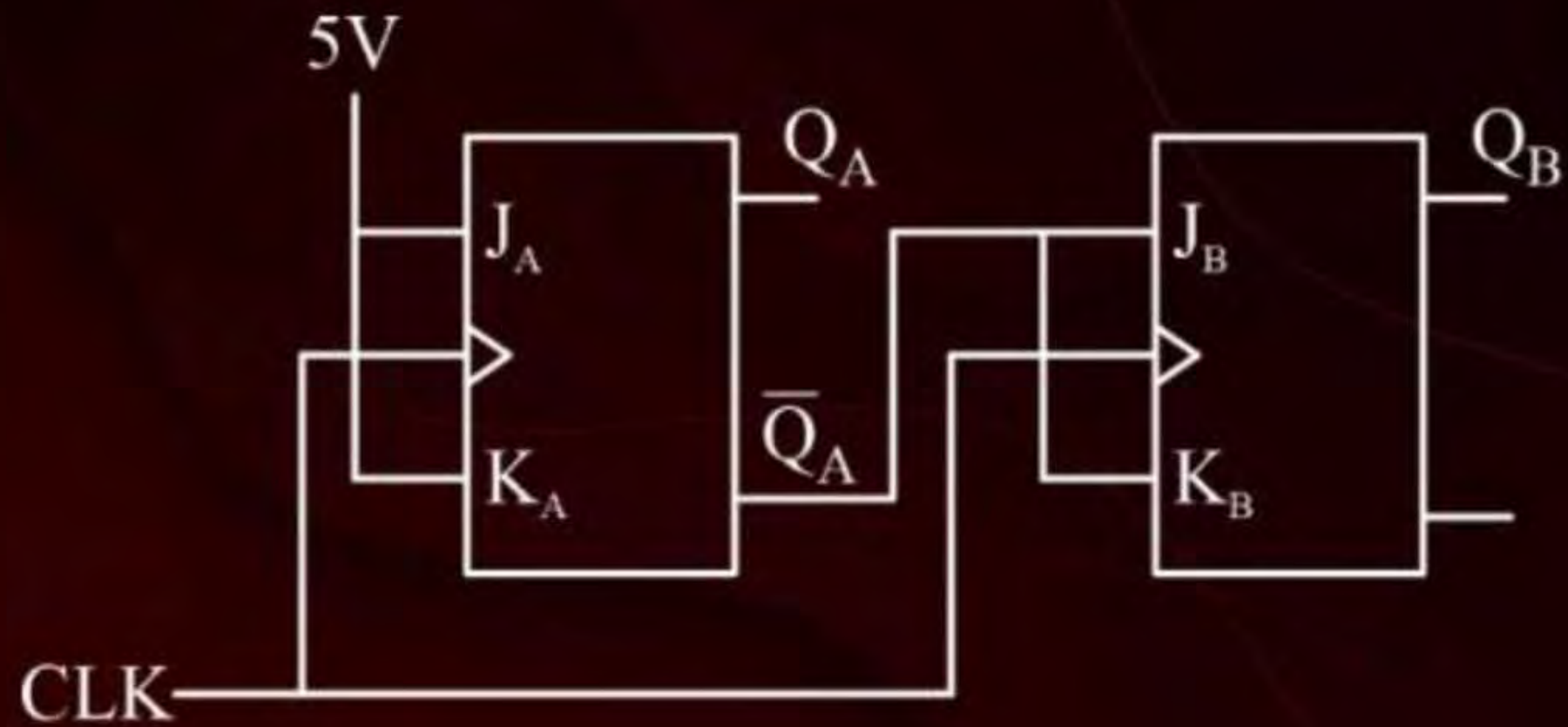
#Q. Which one of the following statements is true about the digital circuit shown in the figure?



- A** It can be used for dividing the input frequency by 3.
- B** It can be used for dividing input frequency by 5.
- C** It can be used for dividing the input frequency by 7.
- D** It cannot be reliably used as a frequency divider due to disjoint internal cycles.

#Q. The current state $Q_A Q_B$ of a two JK flip-flop system is 00. Assume that the clock rise-time is much smaller than the delay of the JK flip-flop. The next state of the system is

- A** 00
- B** 01
- C** 11
- D** 10



#Q. In the following sequential circuit, the initial state (before the first clock pulse) of the circuit is $Q_1Q_0 = 00$. The state $(Q_1 Q_0)$, immediately after the 333rd clock pulse is

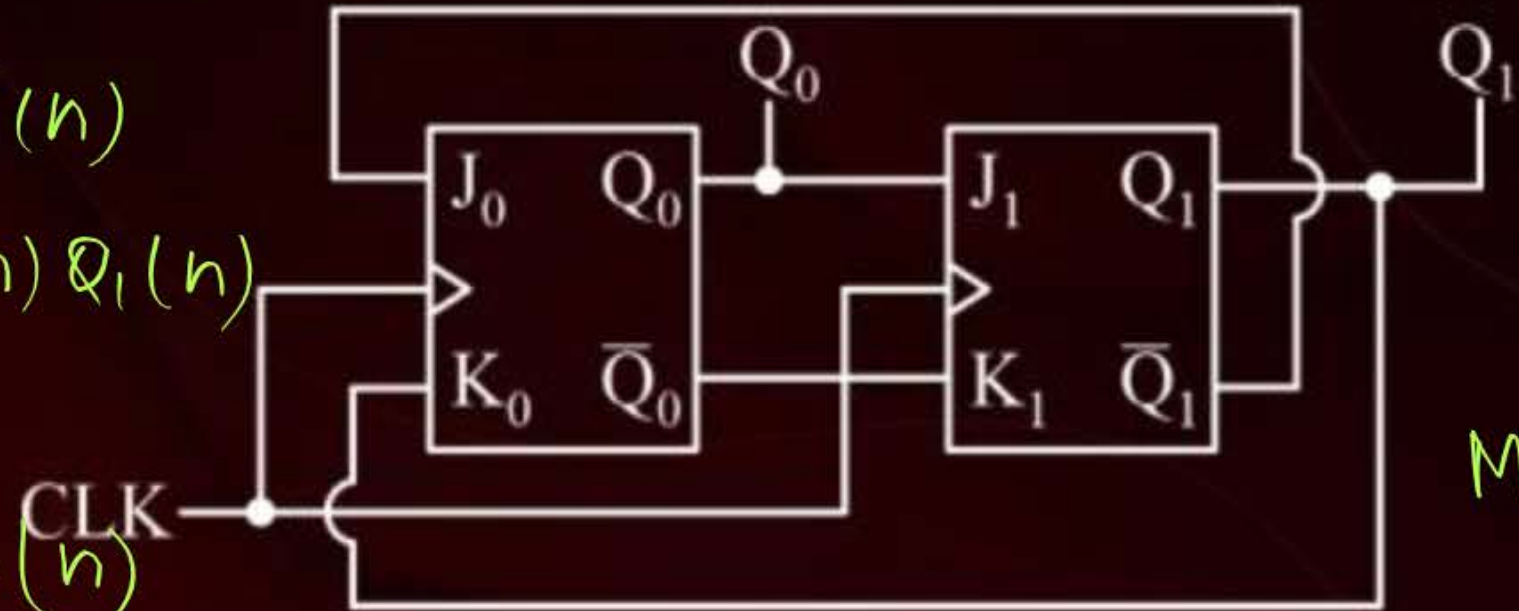
- A 00
- ☒ B 01
- C 10
- D 11

$$Q_1(n+1) = Q_0(n) \bar{Q}_1(n)$$

$$+ Q_0(n) Q_1(n) \\ = Q_0(n)$$

$$Q_0(n+1) = \bar{Q}_1(n) \bar{Q}_0(n)$$

$$+ \bar{Q}_1(n) Q_0(n) \\ = \bar{Q}_1(n)$$



MOD-4

$(0-1-3-2)$
 $(0)_{10} \xrightarrow{332} (0)_{10} \xrightarrow{333^{rd}} (1)_{10} = (01)_2$

	Q_1	Q_0	
1 st	0	0	0 0
2 nd	0	1	
3 rd	1	1	
	1	0	

#Q. A two-bit counter circuit is shown below.

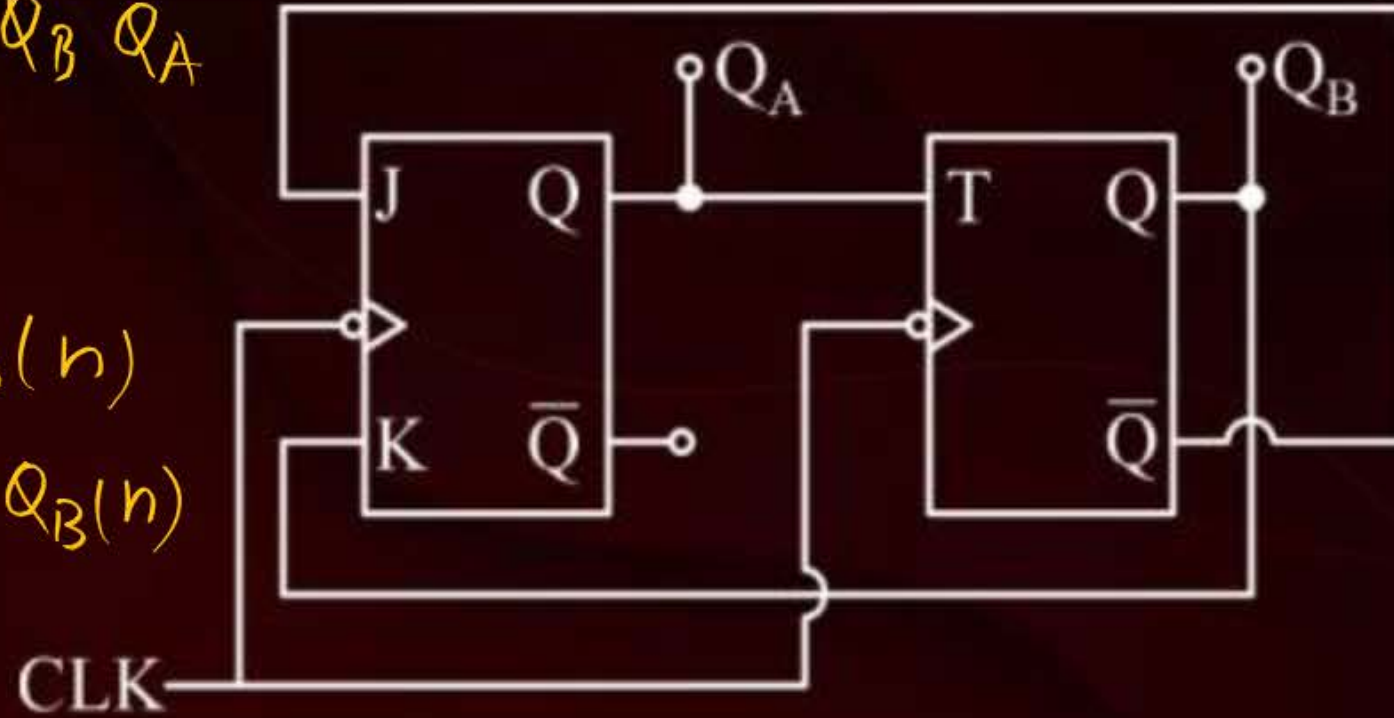
If the state $\underline{Q_A} \underline{Q_B}$ of the counter at the clock time t_n is "10" then the state $Q_A Q_B$ of the counter at $t_n + 3$ (after three clock cycles) will be

$$Q_A(n+1) = \overline{Q_B} \overline{Q_A} + \overline{Q_B} Q_A$$

$$= \overline{Q_B}(n)$$

$$Q_B(n+1) = T \oplus Q_B(n)$$

$$= Q_A(n) \oplus Q_B(n)$$



A 00

B 01

C 10

D 11

$Q_A \quad Q_B$

1 0

1st

1 1

2nd

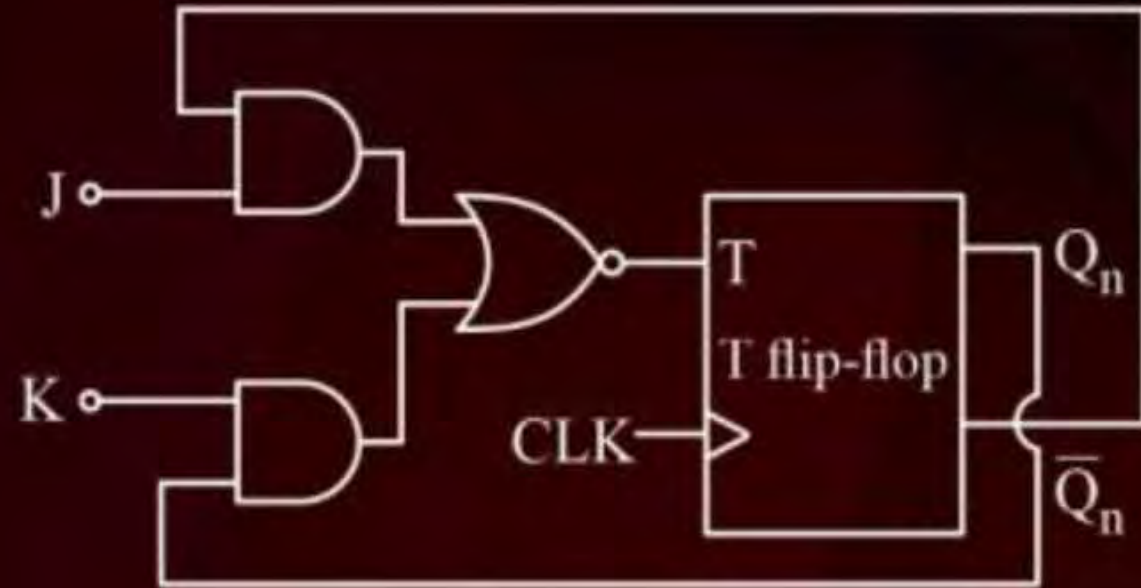
0 0

3rd

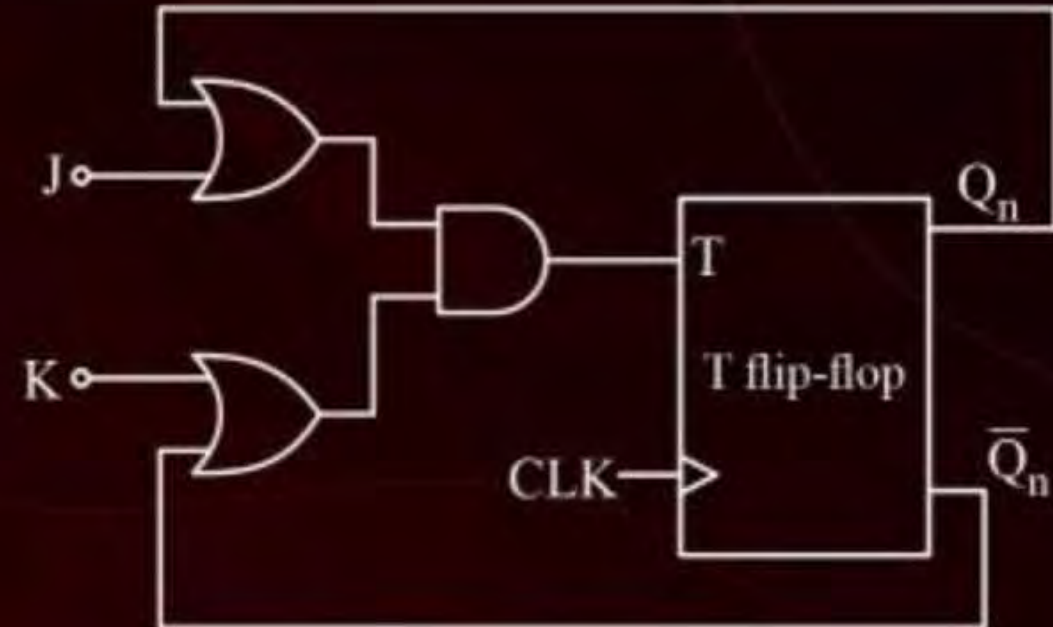
1 0

#Q. A JK flip-flop can be implemented by T flip-flops. Identify the correct implementation.

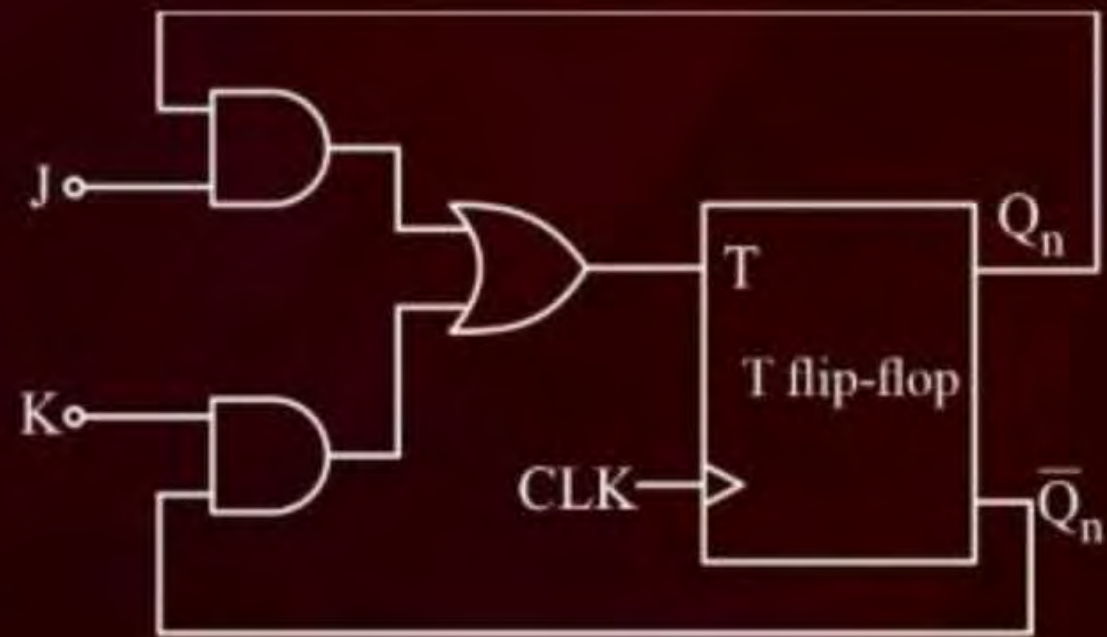
A



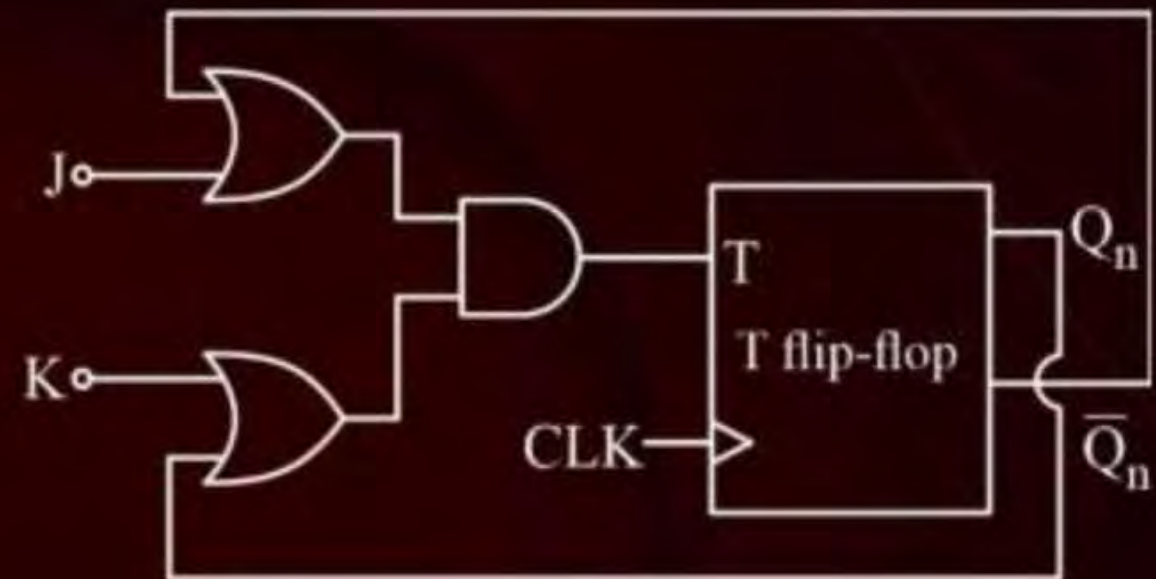
B



C

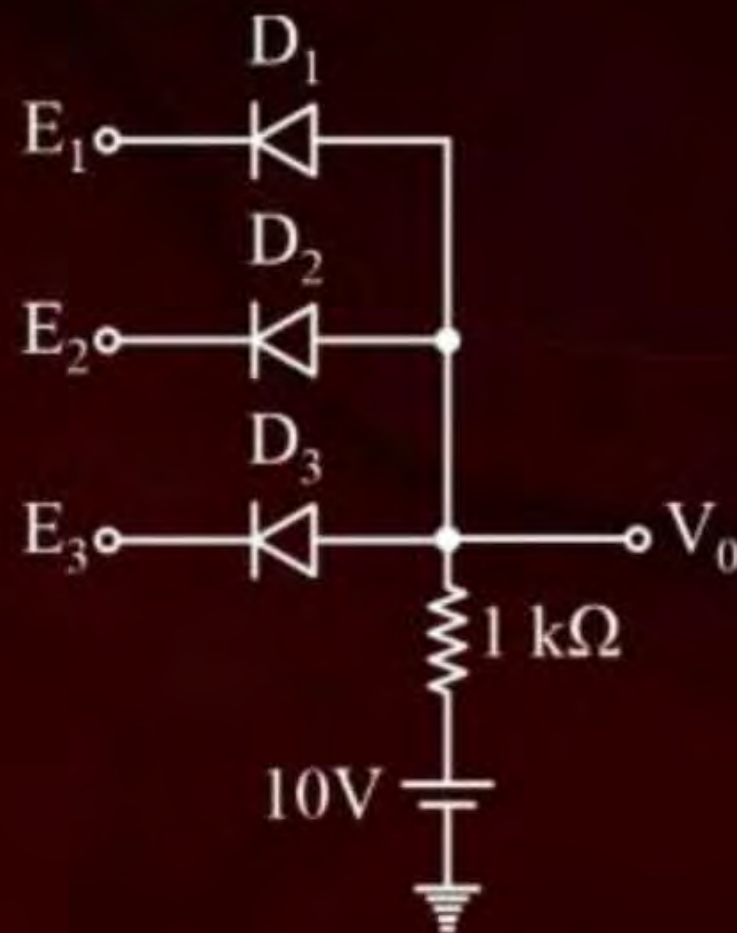


D



#Q. In the circuit shown, diodes D_1 , D_2 and D_3 are ideal, and the inputs E_1 , E_2 and E_3 are '0 V' for logic '0' and '10 V' for logic '1'. What logic gate does the circuit represent?

- A** 3 input OR gate
- B** 3 input NOR gate
- C** 3 input AND gate
- D** 3 input XOR gate



#Q. Assuming that all flip-flops are in reset conditions initially, the count sequence observed at Q_A in the circuit shown is

0 1 1 0 0 0 1 1 0 0 0 $Q_A(n+1) = \overline{Q_B(n)} + Q_C(n)$

$Q_B(n+1) = Q_A(n)$

$Q_C(n+1) = Q_B(n)$

A

0010111...

B

0001011...

C

0101111...

D

0110100...

Q_A Q_B Q_C
0 0 0

1 0 0

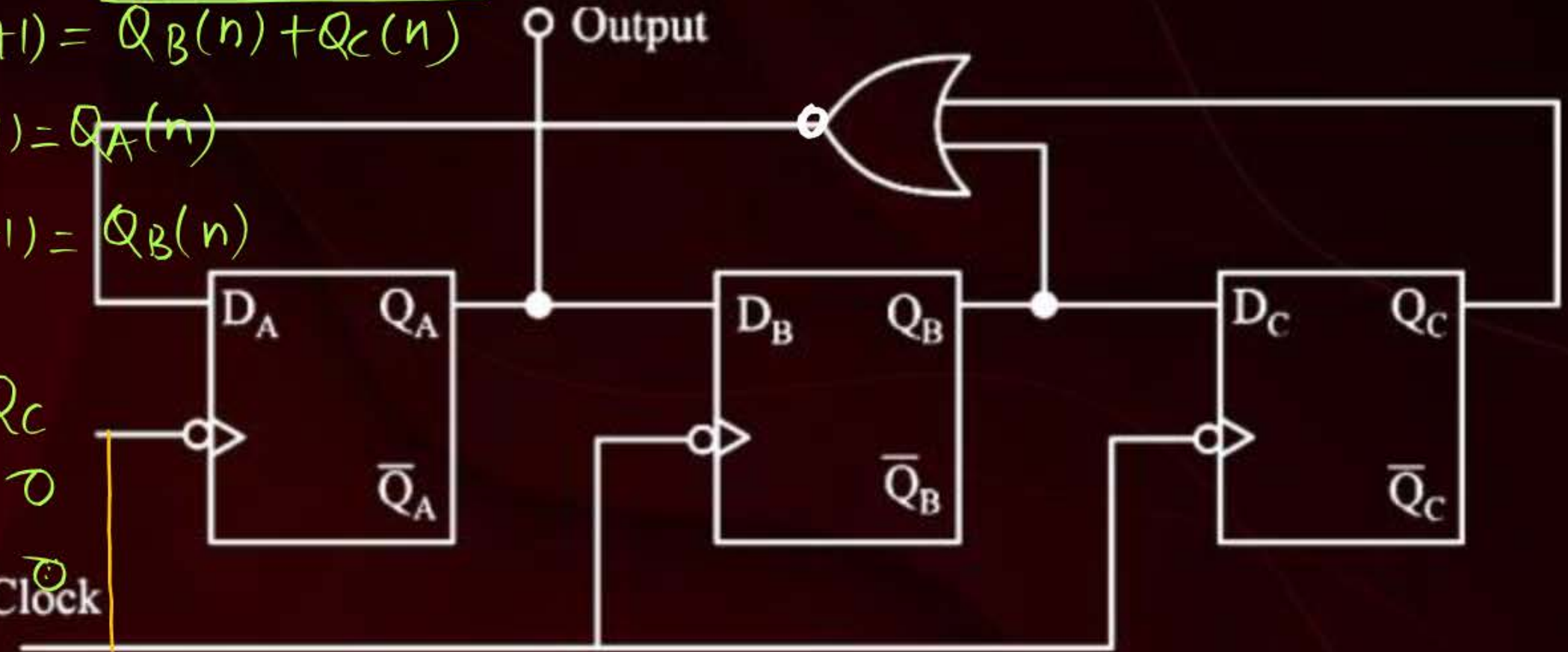
1 1 0

0 1 1

0 0 1

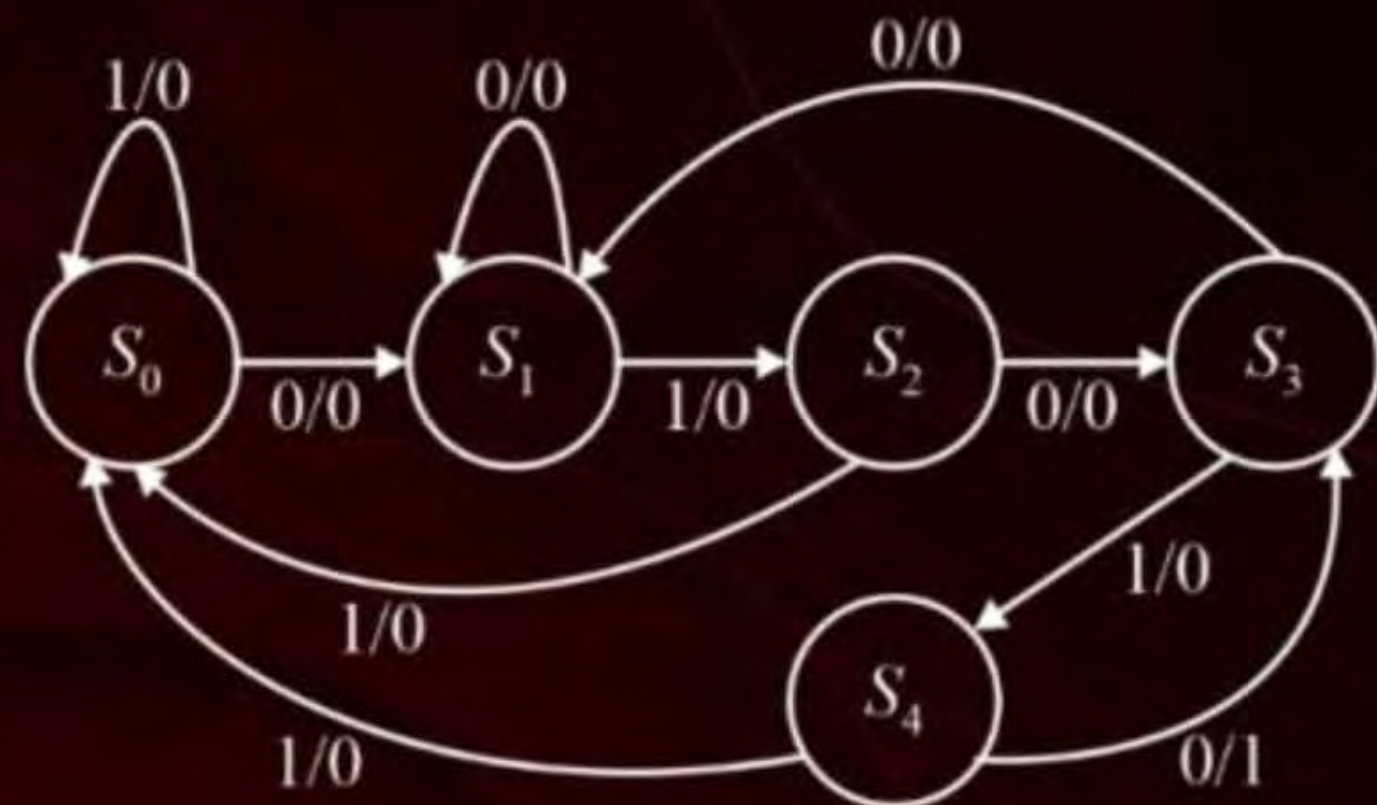
0 0 0

Clock

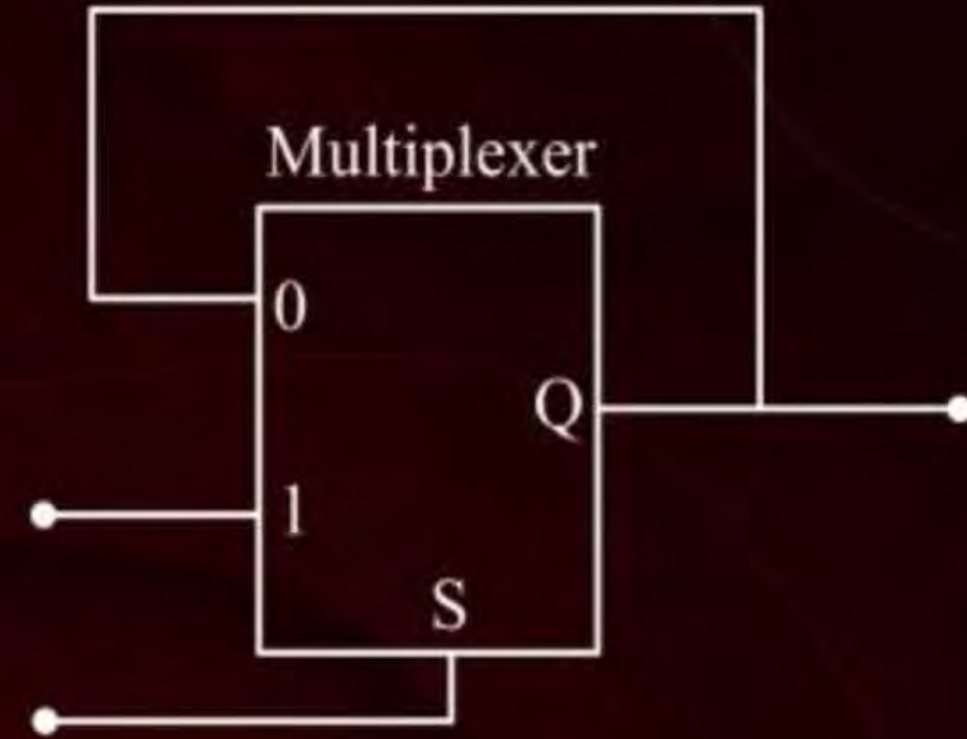


#Q. The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1. Then

- A** the sequence 01010 is detected.
- B** the sequence 01011 is detected.
- C** the sequence 01110 is detected.
- D** the sequence 01001 is detected.

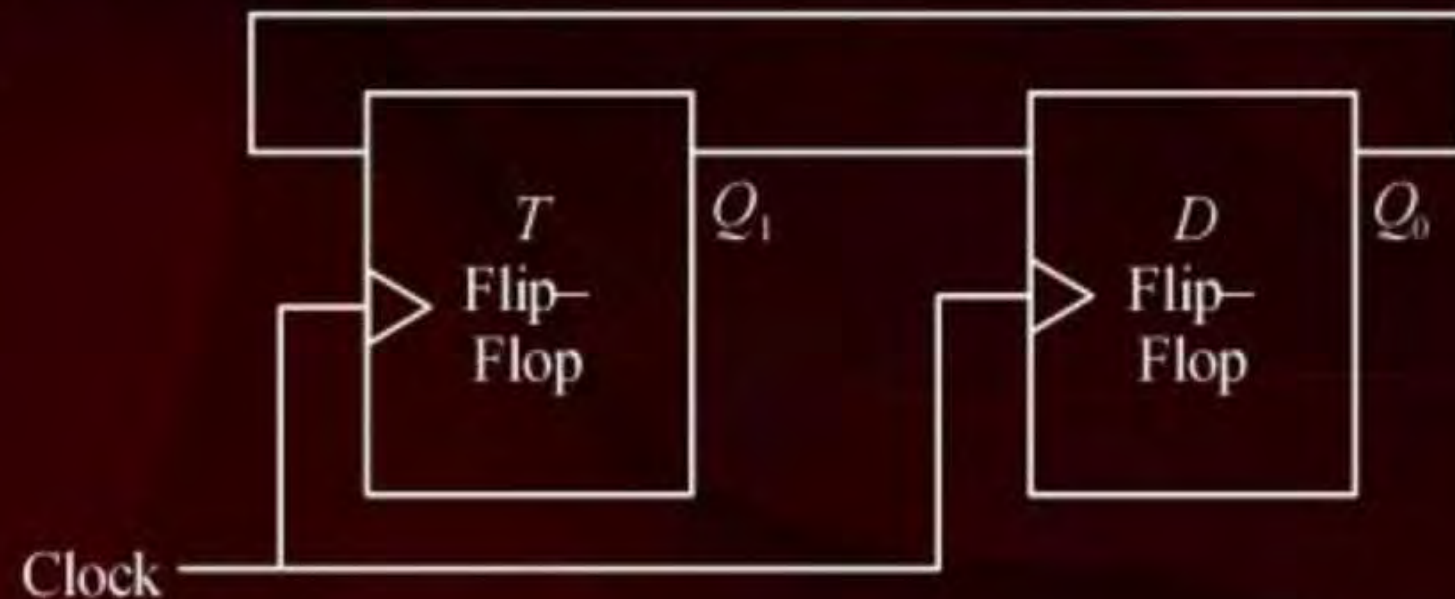


#Q. The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.
Match the function equivalence of this circuit to one of the following options.



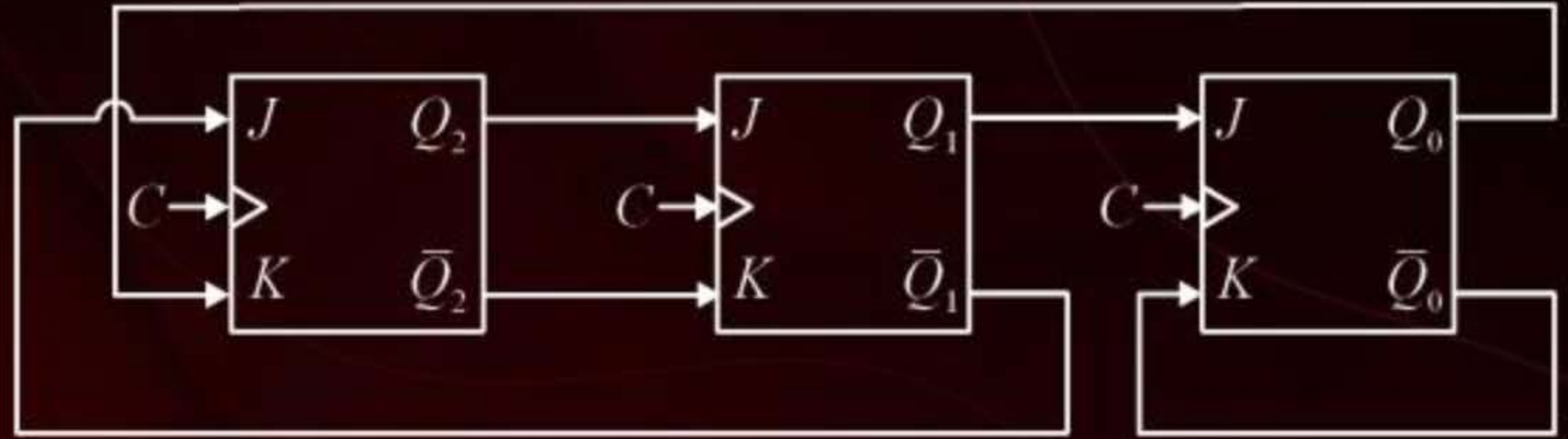
- A** D Flip-flop
- B** D Latch
- C** Half-adder
- D** Demultiplexer

#Q. Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



- A** Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
- B** Q_1Q_0 after the 3rd cycle are 11 and after the 4th cycle are 01 respectively
- C** Q_1Q_0 after the 3rd cycle are 00 and after the 4th cycle are 11 respectively
- D** Q_1Q_0 after the 3rd cycle are 01 and after the 4th cycle are 01 respectively.

#Q. The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is



A 001, 010, 011 ✗

B 111, 110, 101 ✗

C 100, 110, 111 ✓

D 100, 011, 001 ✗

$Q_2 Q_1 Q_0$
0 0 0

1 0 0 ✓

1st ✓

2nd ✓ 1

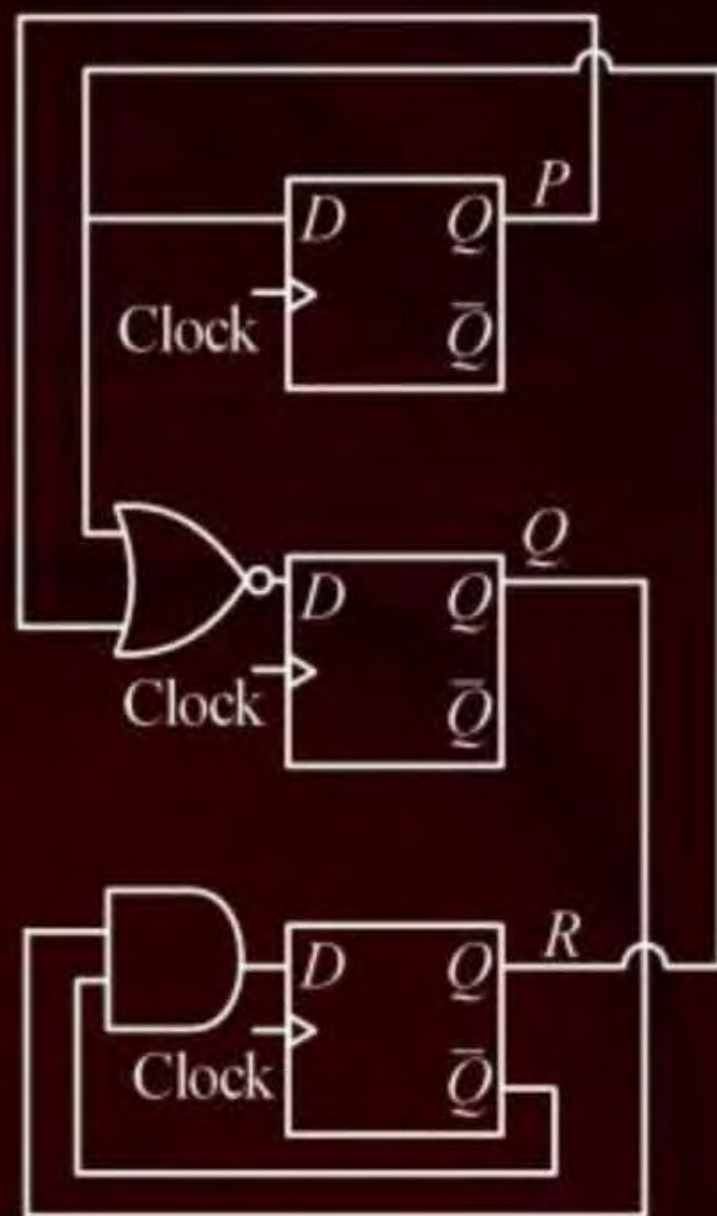
$$Q_2(n+1) = \overline{Q_1(n)} \overline{Q_2(n)} + \overline{Q_0(n)} Q_2(n)$$

$$Q_1(n+1) = Q_2(n) \overline{Q_1(n)} + Q_2(n) \cdot Q_1(n) = Q_2(n)$$

$$Q_0(n+1) = Q_1(n) \overline{Q_0(n)} + Q_0(n) \cdot Q_0(n) = Q_0(n) + (Q_1(n) \overline{Q_0(n)}) = Q_0(n) + Q_1(n)$$

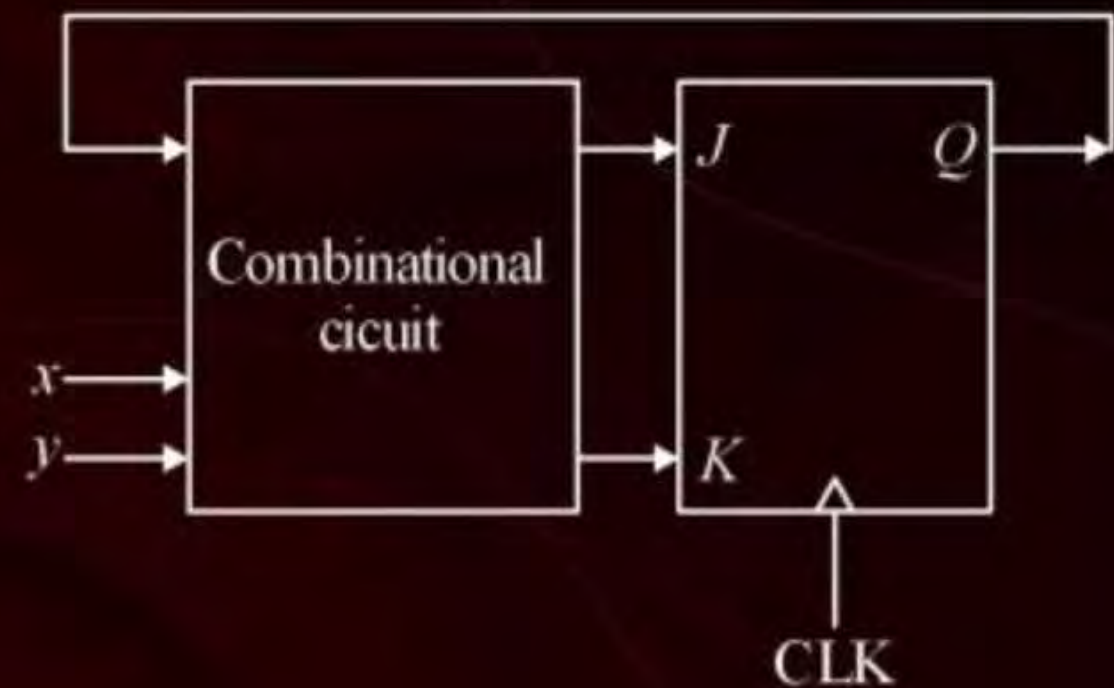
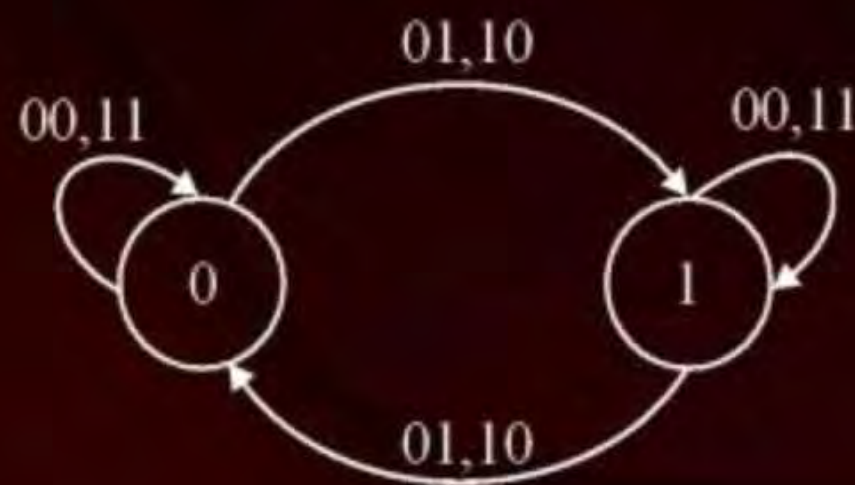
#Q. Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter?

- | | |
|---|---|
| A | 3 |
| B | 4 |
| C | 5 |
| D | 6 |

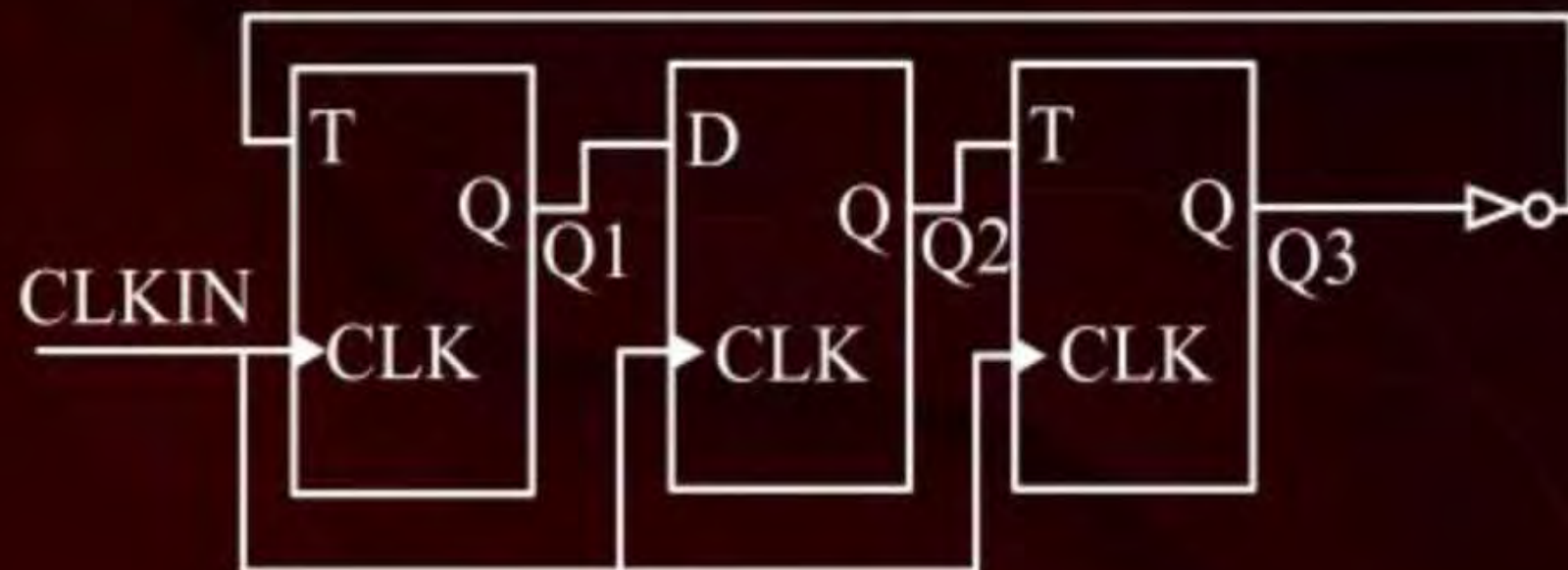


#Q. Consider the following state diagram and its realization by a JK flip flop
The combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are:

- A** $(x \oplus y)'$ and $(x \oplus y)'$
- B** $(x \oplus y)'$ and $x \oplus y$
- C** $x \oplus y$ and $(x \oplus y)'$
- D** $x \oplus y$ and $x \oplus y$



#Q. Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively. Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?



A (0, 0, 1)

B (1, 0, 0)

C (1, 0, 1)

D (1, 1, 1)

#Q. Let $k = 2^n$. A circuit is built by giving the output of an n -bit binary counter as input to an n -to- 2^n bit decoder. This circuit is equivalent to a

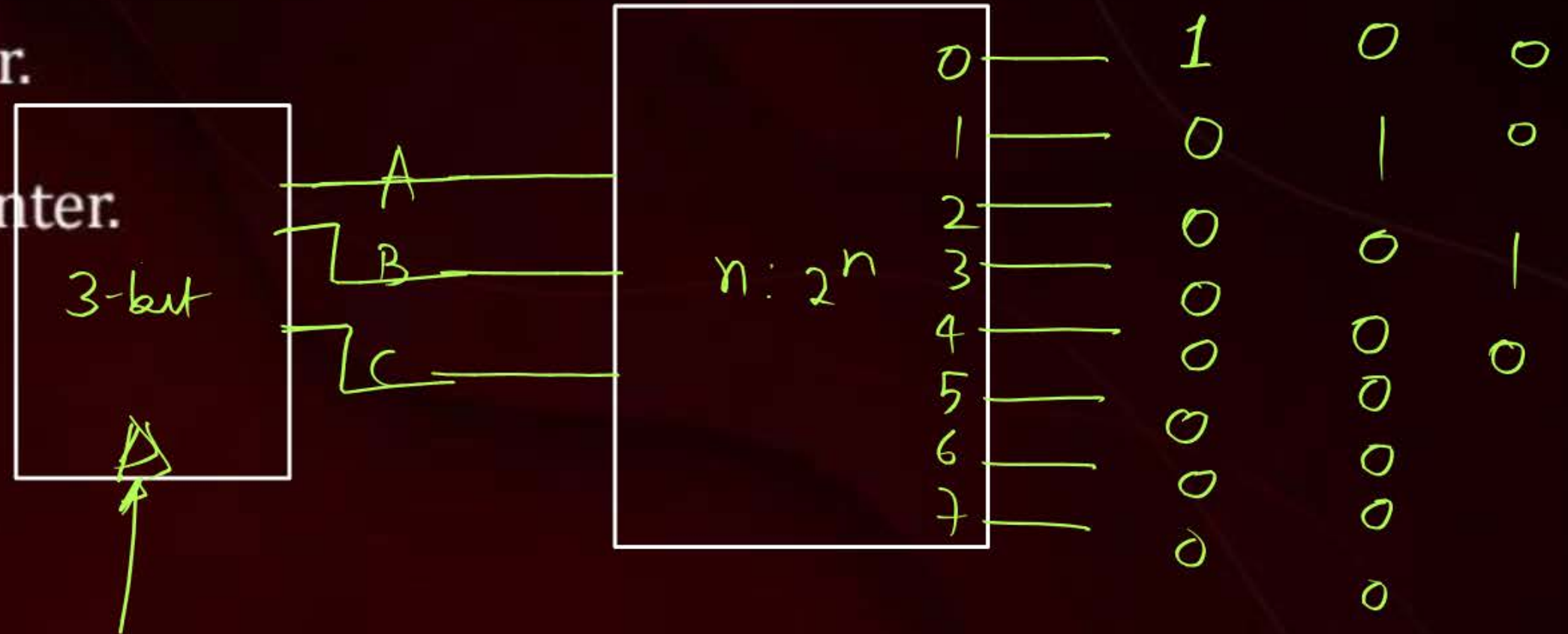
$\checkmark k = 8$ $0 - (2^k - 1)$

A k -bit binary up counter.

B k -bit binary down counter.

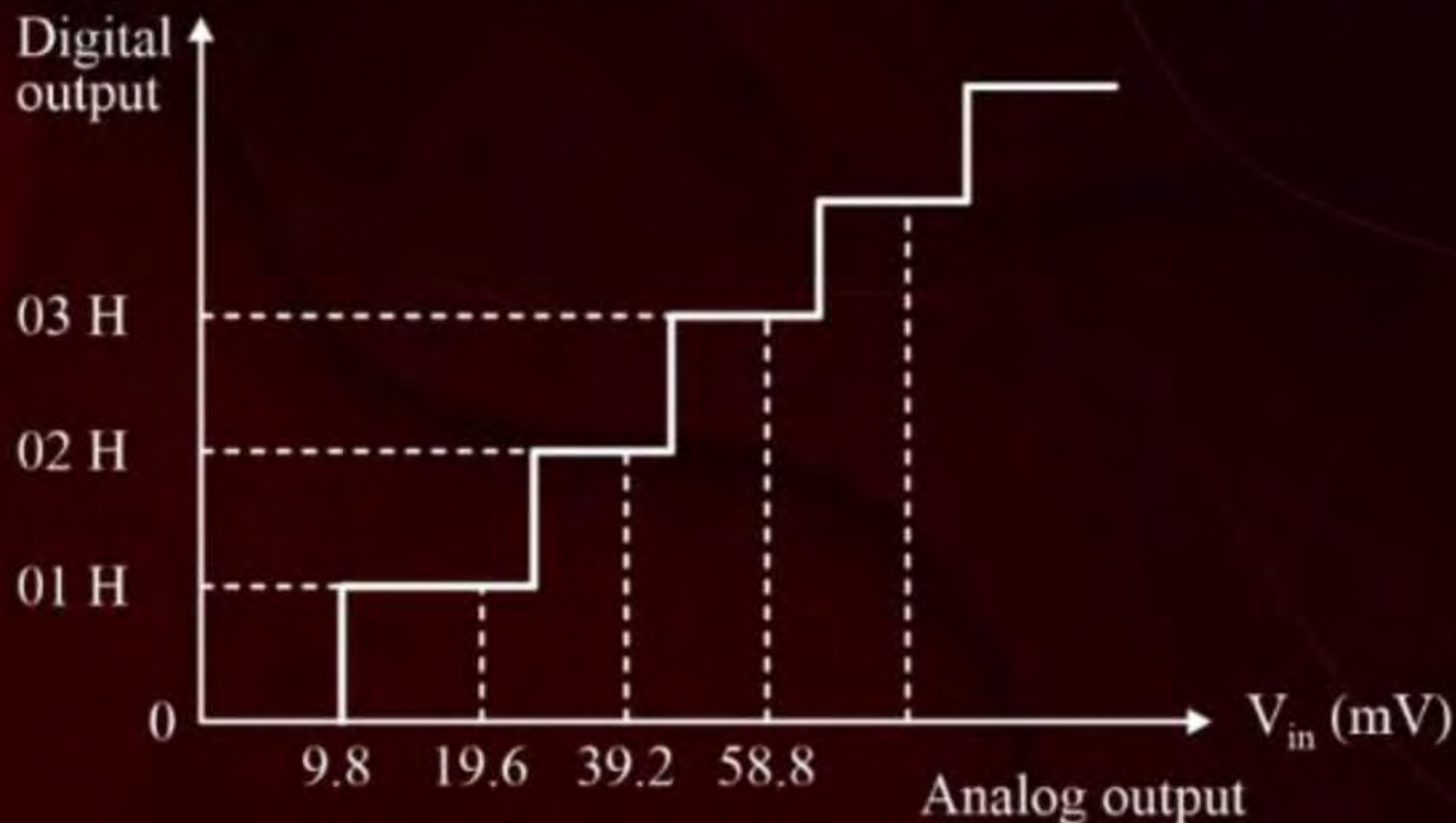
C k -bit ring counter.

D k -bit Johnson counter.



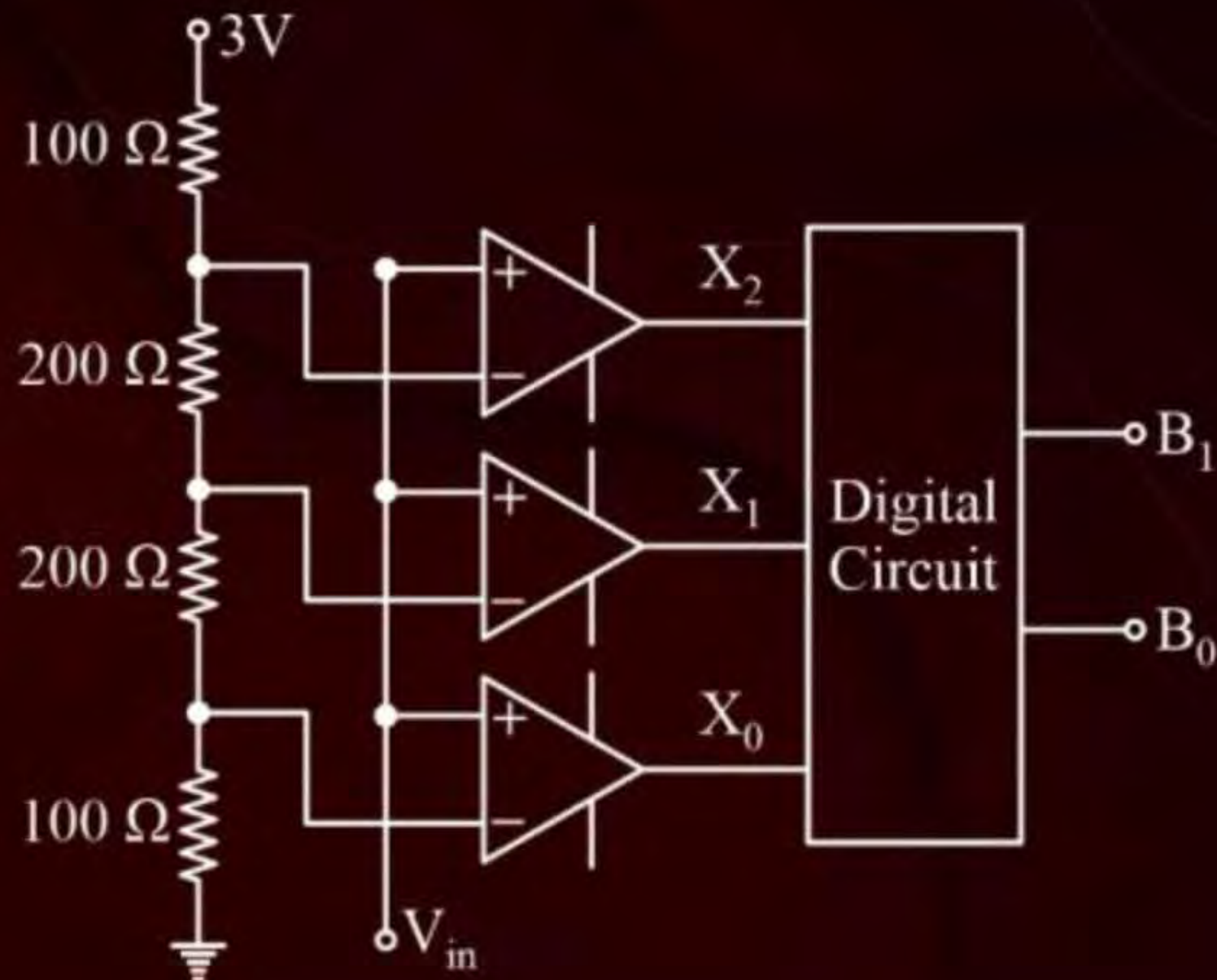
#Q. An 8 bit ADC converts analog voltage in the range of 0 to +5 V to the corresponding digital code as per the conversion characteristics shown in figure. For $V_{in} = 1.9922 \text{ V}$, which of the following digital output, given in hex, is true ?

- A** 64H
- B** 66H
- C** 65H
- D** 67H



#Q. A 2-bit flash Analog to Digital Converter (ADC) is given below. The input is $0 < V_{IN} < 3$ volts. The expression for the LSB of the output B_0 as a Boolean function of X_2 , X_1 and X_0 is

- A** $X_0 [X_2 \oplus X_1]$
- B** $\bar{X}_0 [X_2 \oplus X_1]$
- C** $X_0 [X_2 \oplus X_1]$
- D** $\bar{X}_0 [X_2 \oplus X_1]$



#Q. A temperature in the range of -40°C to 55°C is to be measured with a resolution of 0.1°C . The minimum number of ADC bits required to get a matching dynamic range of the temperature sensor is

- A** 8
- B** 10
- C** 12
- D** 14

#Q. An 8-bit, unipolar Successive Approximation Register type ADC is used to convert 3.5 V to digital equivalent output. The reference voltage is +5 V. The output of the ADC at the end of 3rd clock pulse after the start of conversion, is

- A** 10100000
- B** 10000000
- C** 00000001
- D** 00000011

#Q. Two 8-bit ADCs, one of single slope integrating type and other of successive approximation type, take T_A and T_B times to convert 5 V analog input signal to equivalent digital output. If the input analog signal is reduced to 2.5V. The approximate time taken by the two ADCs will respectively, be

- A** T_A, T_B
- B** $T_A/2, T_B$
- C** $T_A, T_B/2$
- D** $T_A/2, T_B/2$

#Q. The signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input is given to be 61.96 dB. The resolution of the ADC is _____ bits (rounded off to the nearest integer).

- A** 10
- B** 100
- C** 200
- D** 1024

#Q. An 8-bit unipolar (all analog output values are positive) digital-to-analog converter (DAC) has a full-scale voltage range from 0 V to 7.68 V. If the digital input code is 10010110 (the leftmost bit is MSB), then the analog output voltage of the DAC (rounded off to one decimal place) is ____ V.

- A** 1.5
- B** 3.5
- C** 4.5
- D** 5.5

#Q. A 10-bit D/A converter is calibrated over the full range from 0 to 10 V. If the input to the D/A converter is 13 A (in hex), the output (rounded off to three decimal places) is_____V.

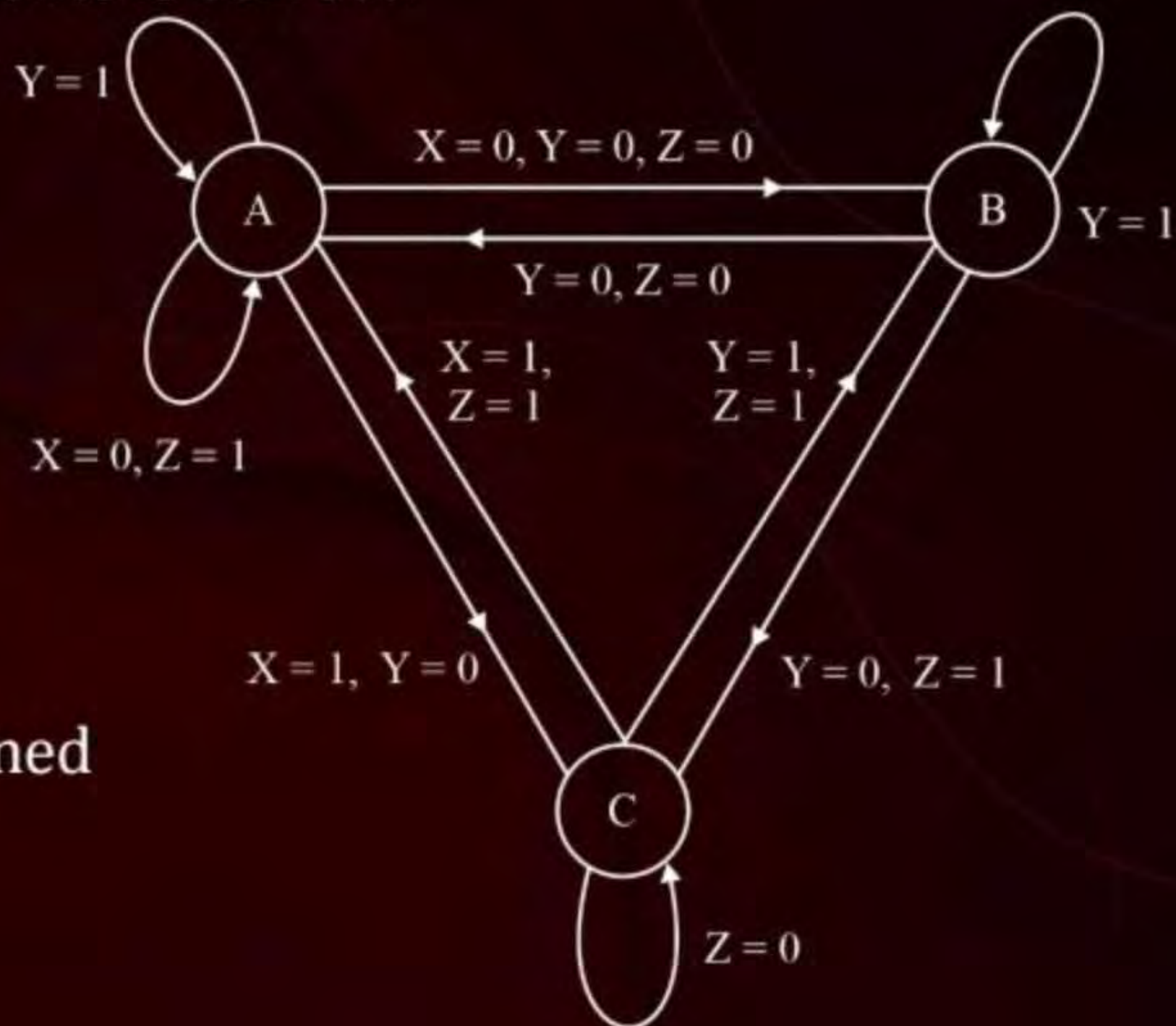
- A** 2.66
- B** 1.33
- C** 4.07
- D** 3.07

#Q. A sequence detector is designed to detect precisely 3 digital inputs, with overlapping sequences detectable. For the sequence (1,0,1) and input data (1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 0), what is the output of this detector?

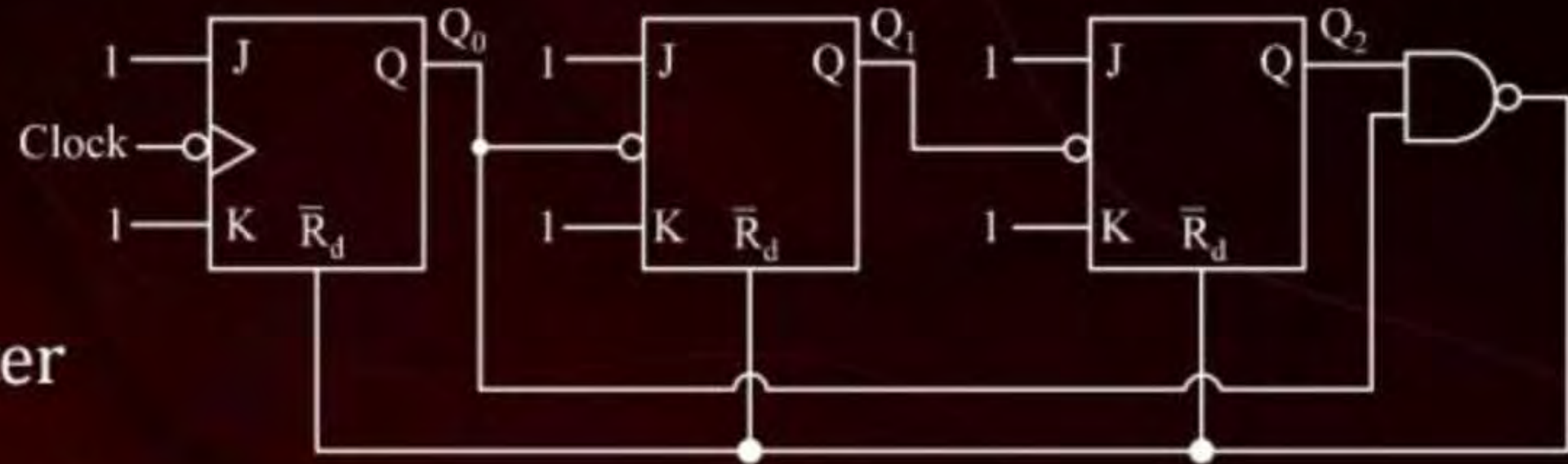
- A** 1,1,0,0,0,0,1,1,0,1,0,0
- B** 0,1,0,0,0,0,0,1,0,1,0,0
- C** 0,1,0,0,0,0,0,1,0,1,1,0
- D** 0,1,0,0,0,0,0,0,1,0,0,0

#Q. The state transition diagram for a finite state machine with states A, B and C, and binary inputs X, Y and Z, is shown in the figure. Which one of the following statements is correct?

- A** Transitions from State A are ambiguously defined.
- B** Transitions from State B are ambiguously defined.
- C** Transitions from State C are ambiguously defined.
- D** All of the state transitions are defined unambiguously.



#Q. The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input). The counter corresponding to this circuit is

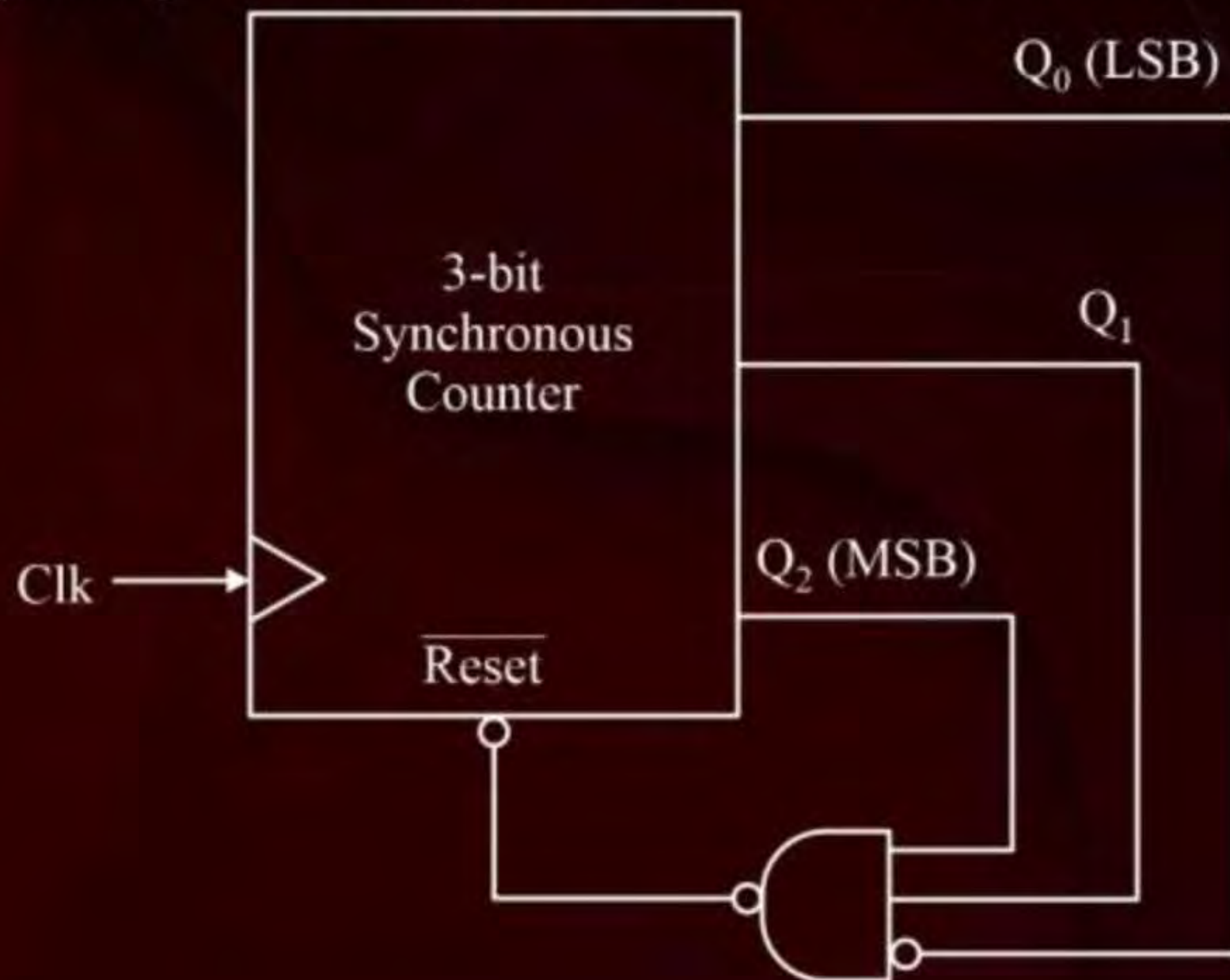


- A** a modulo-5 binary up counter
- B** a modulo-6 binary down counter
- C** a modulo-5 binary down counter
- D** a modulo-6 binary up counter

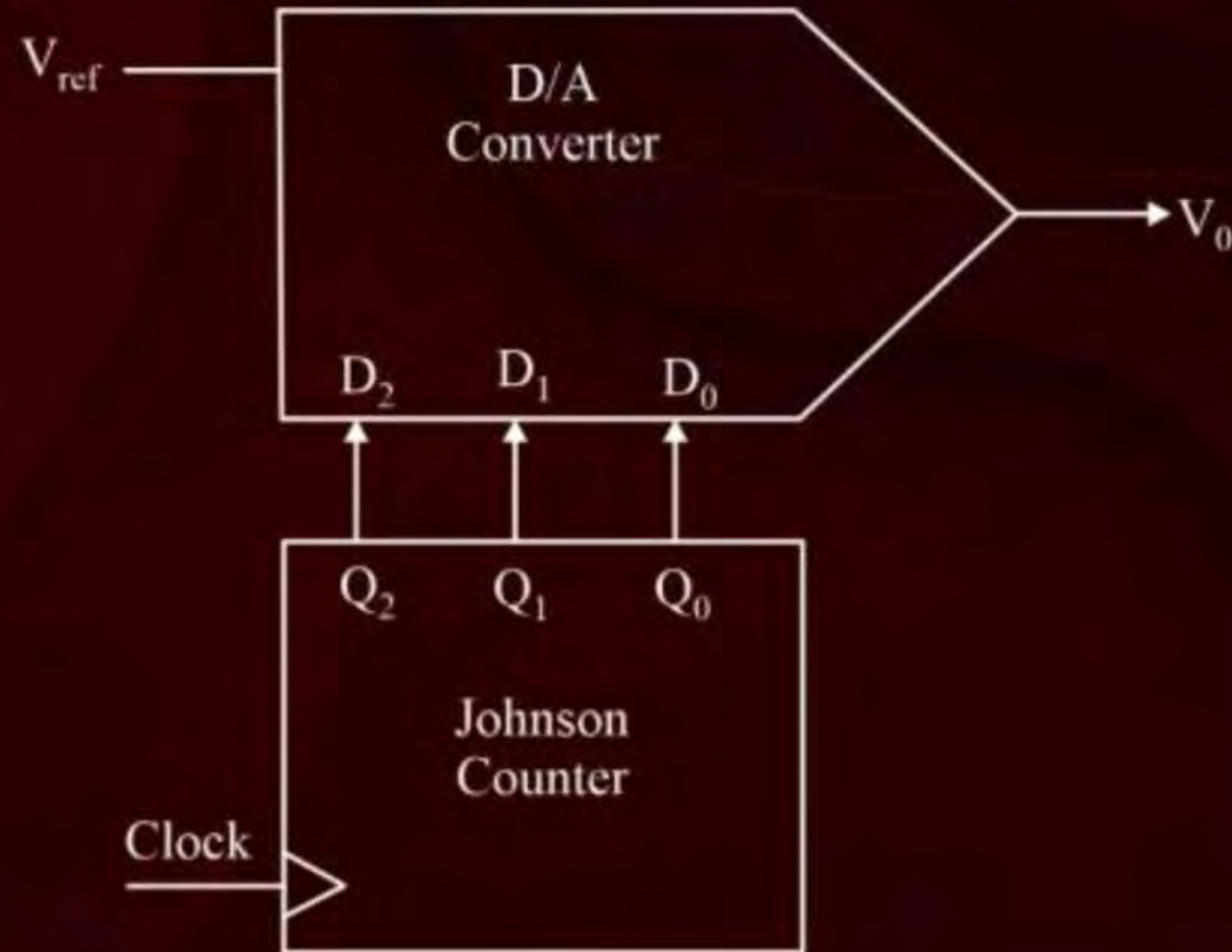
#Q. For the circuit shown in the figure, the delay of the bubbled NAND gate is 2 ns and that of the counter is assumed to be zero.

If the clock (CLK) frequency is 1 GHz, then the counter behaves as a

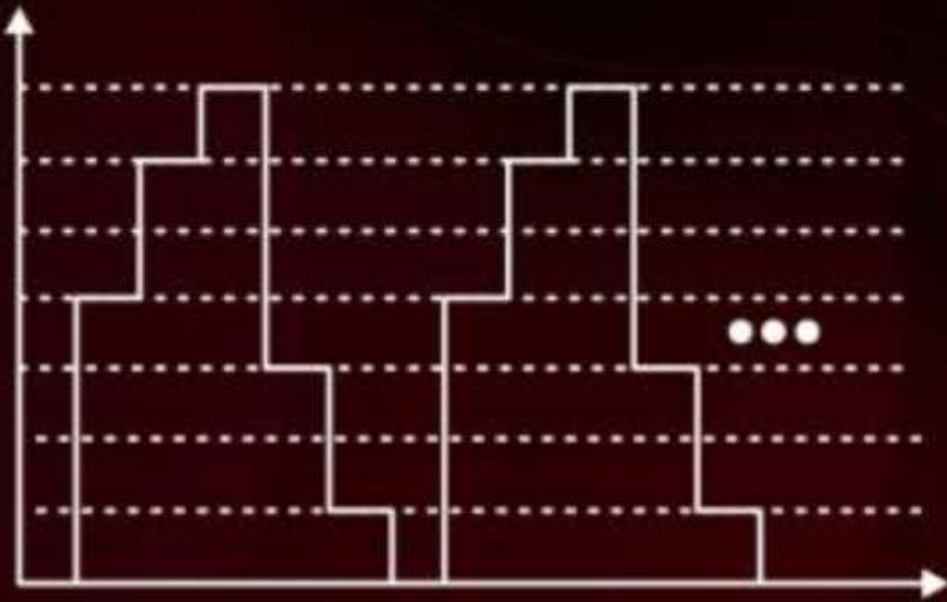
- A** mod-5 counter
- B** mod-6 counter
- C** mod-7 counter
- D** mod-8 counter



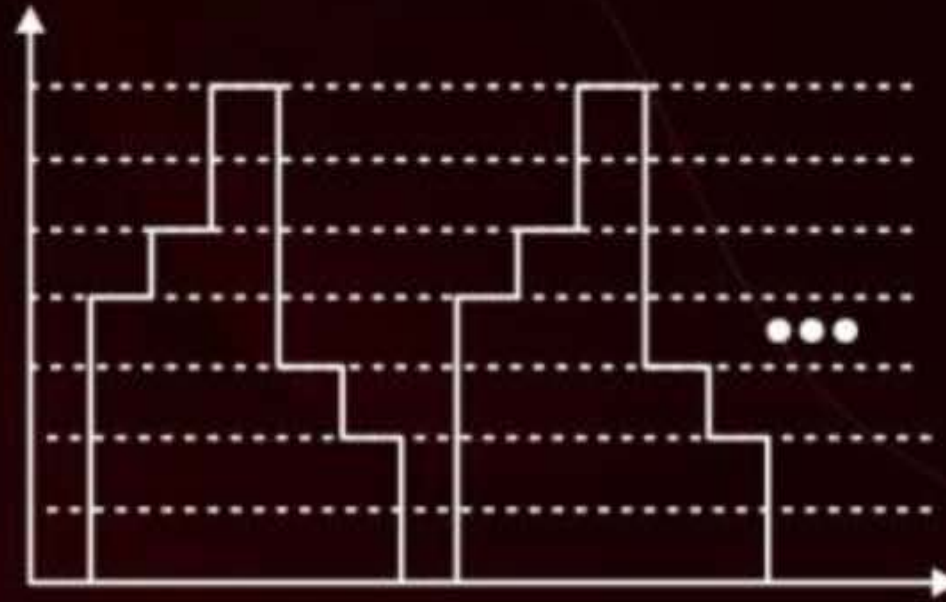
#Q. The output of a 3-stage Johnson (twisted-ring) counter is fed to a digital-to-analog (D/A) converter as shown in the figure below. Assume all states of the counter to be unset initially. The waveform which represents the D/A converter output V_0 is



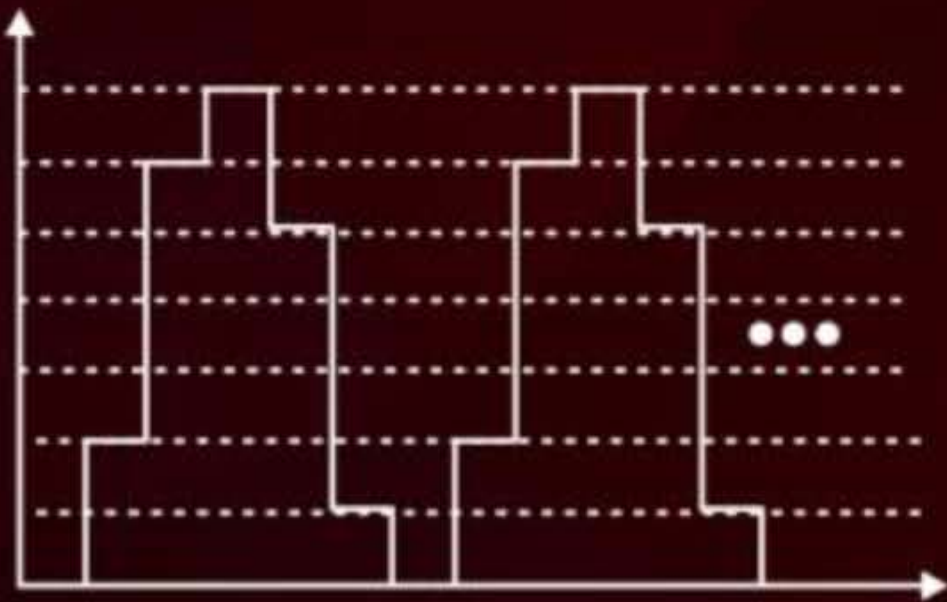
A



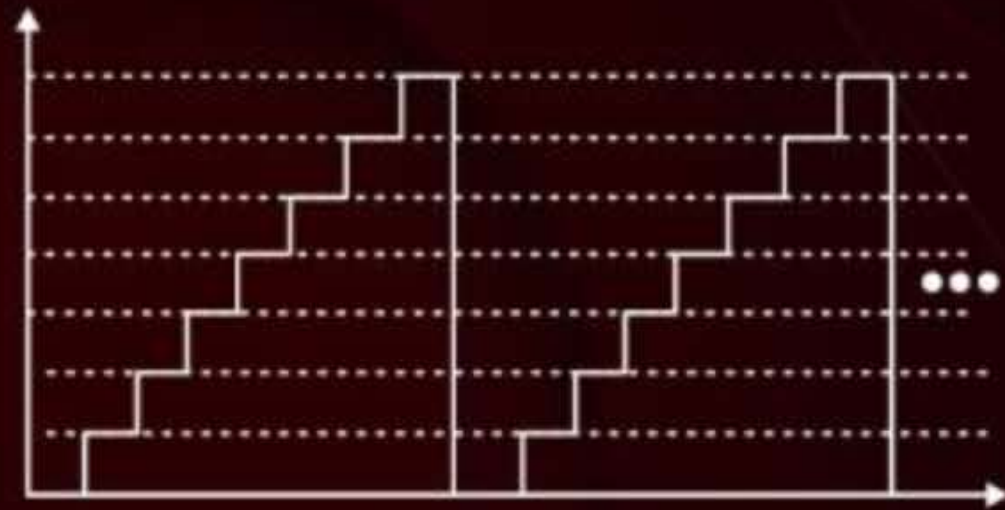
B



D



C



#Q. Consider a four-bit D to A converter. The analog value corresponding to a digital signal of values 0000 and 0001 are 0 V and 0.0625 V respectively. The analog value (in Volts) corresponding to the digital signal 1111 is _____.

- A** 0.94
- B** 0.125
- C** 0.67
- D** None of these

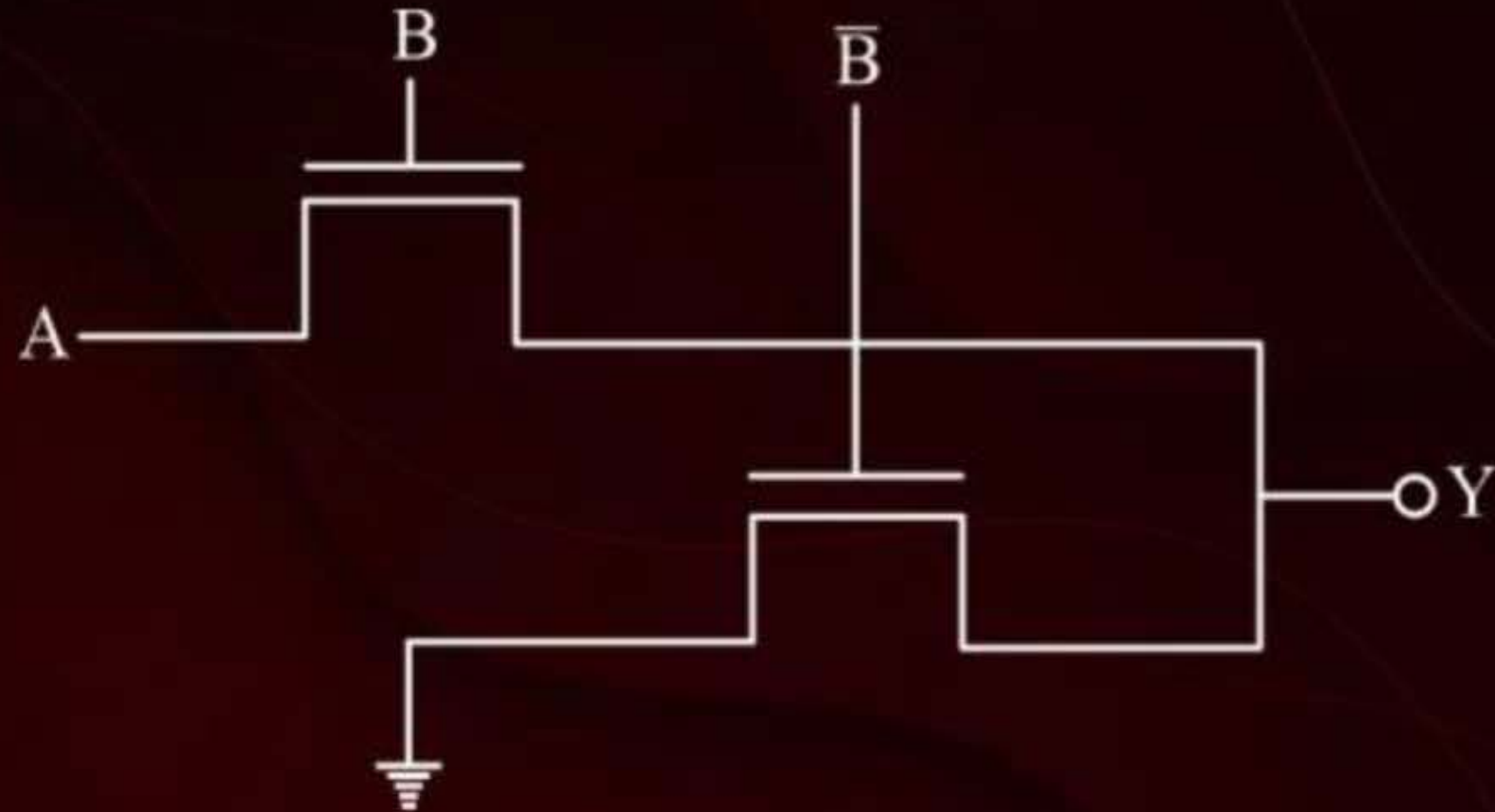
#Q. For the circuit shown in the figure, P and Q are the inputs and Y is the output
The logic implemented by the circuit is

- A** XNOR
- B** XOR
- C** NOR
- D** OR



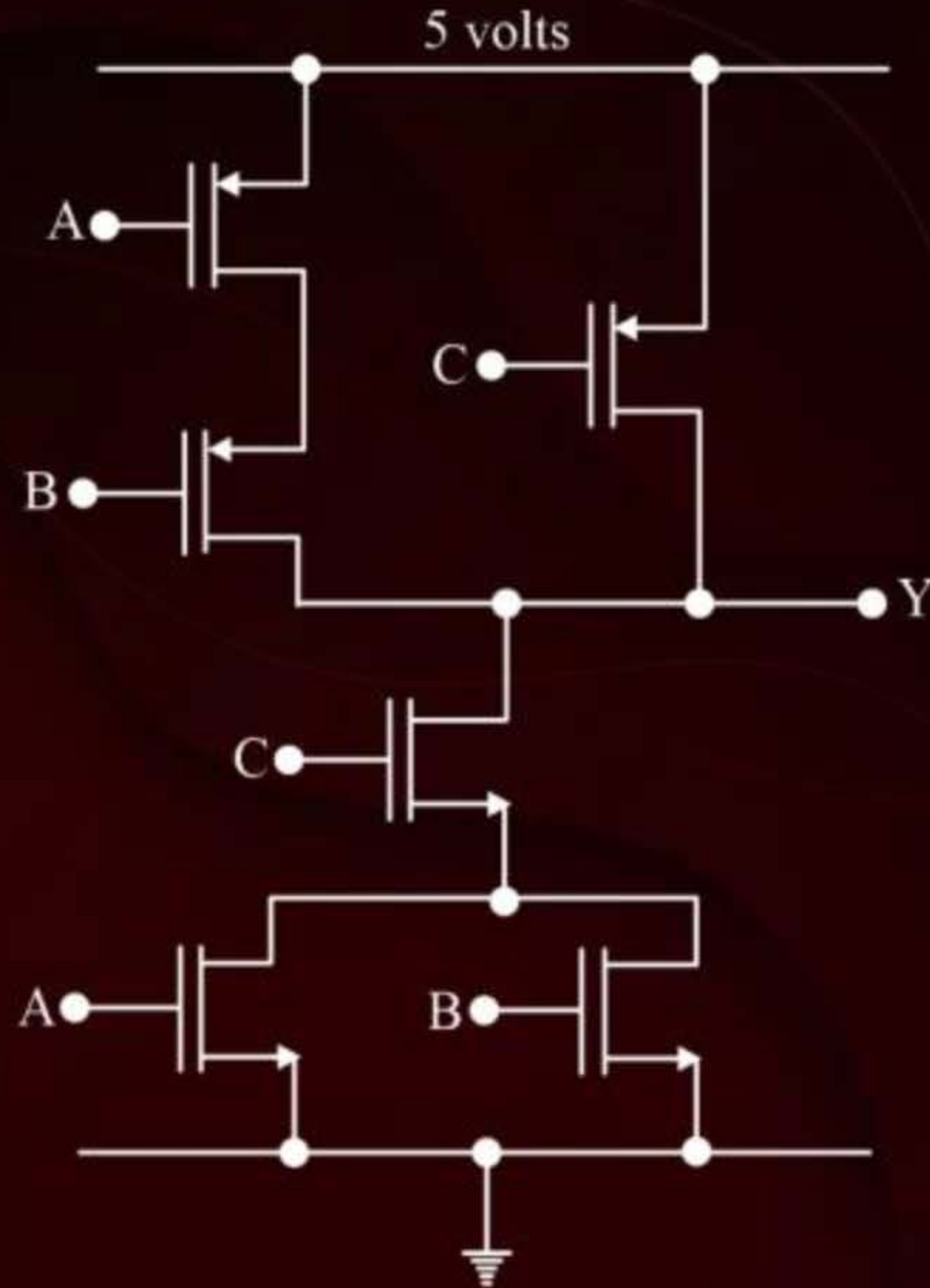
#Q. The logic functionality realized by the circuit shown below is

- A** OR
- B** XOR
- C** NAND
- D** AND



#Q. In the circuit shown

- A** $Y = \bar{A}\bar{B} + \bar{C}$
- B** $Y = (A + B)C$
- C** $Y = (\bar{A} + \bar{B})\bar{C}$
- D** $Y = AB + C$





*Thank
You*